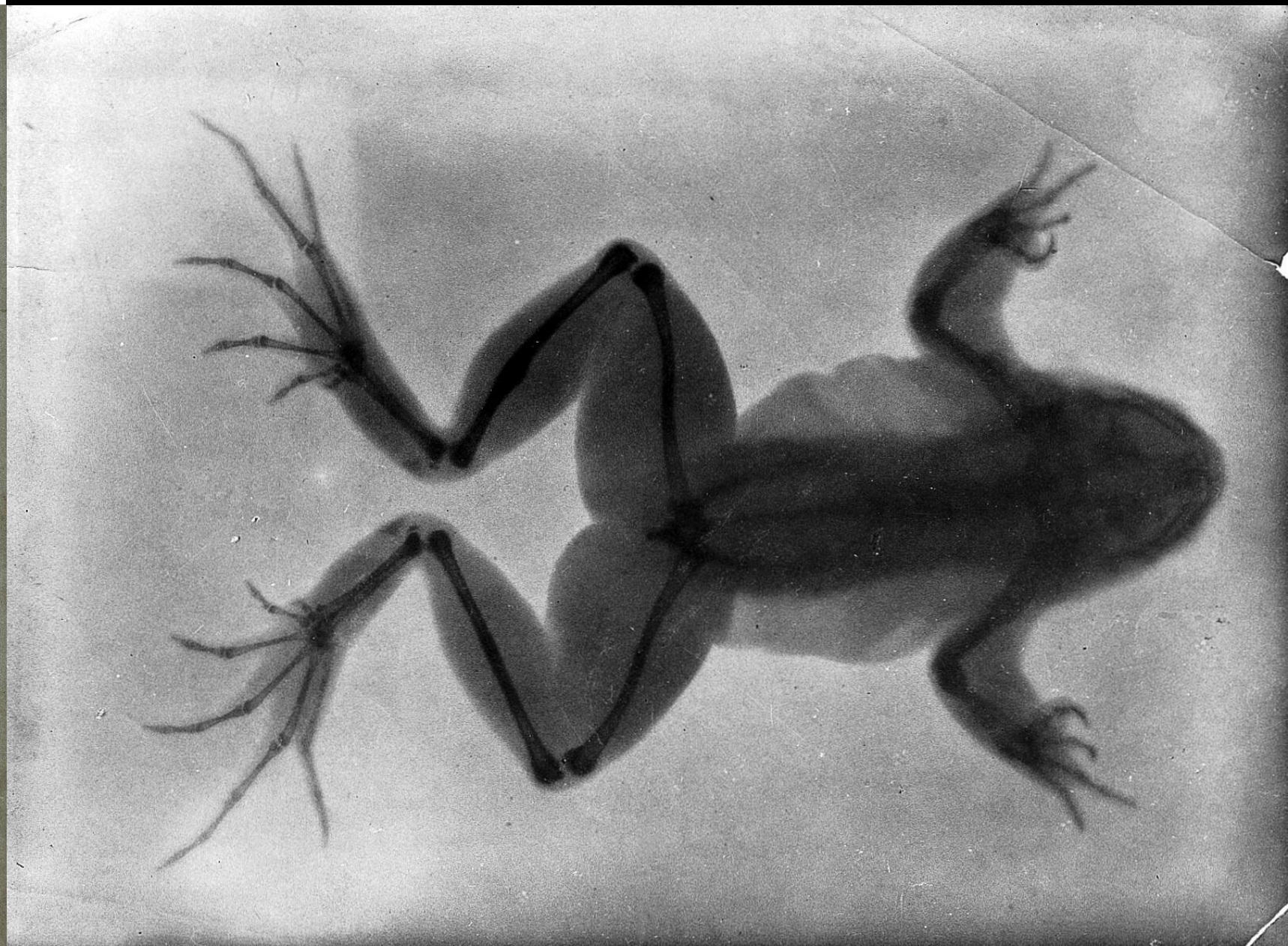
X-ray diffraction

Left: The bones of a hand with a ring on one finger, viewed through X-ray. Photoprint from radiograph by W.K. Röntgen, 1895. Attribution-Non Commercial 4.0 International (CC BY-NC 4.0).
Right: Radiograph: frog with a broken leg that has healed, by A. Schuster, 1896. Credit: Wellcome Collection. Attribution 4.0 International (CC BY 4.0).

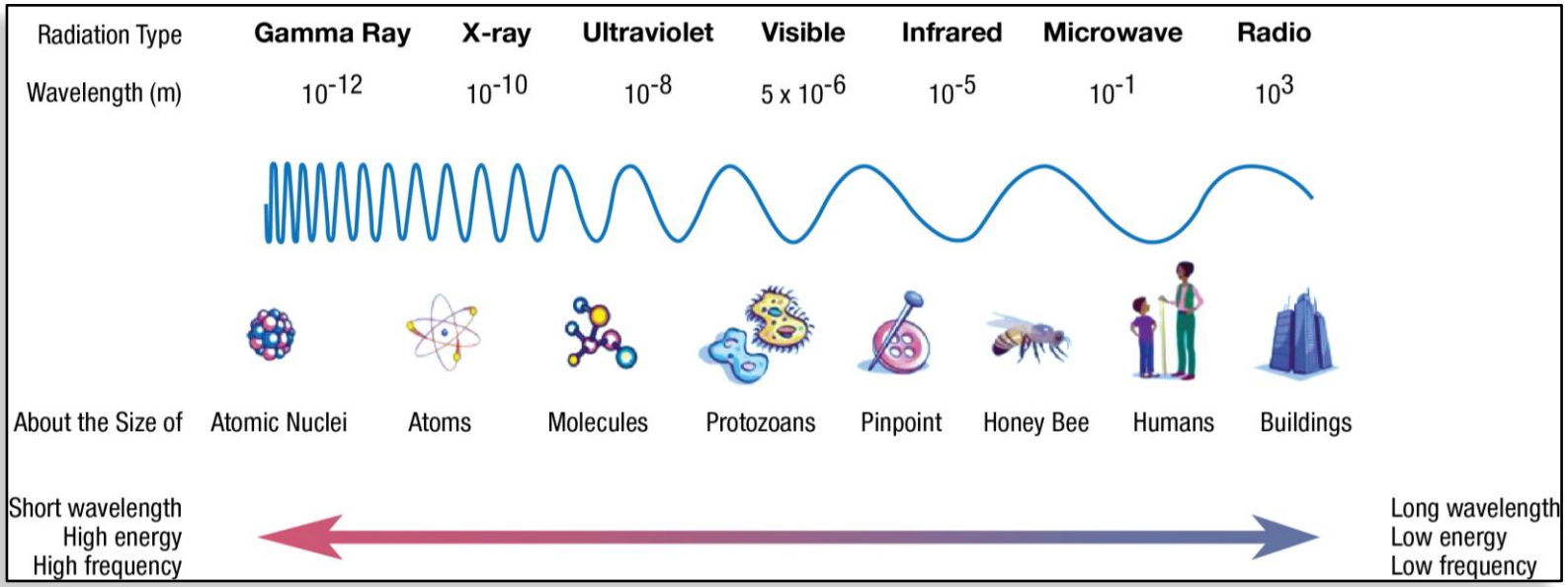




What are X-rays?

X-ray is a penetrating form of **high-energy** electromagnetic radiation.

Most X-rays have wavelengths (λ) of 10^{-12}–10^{-8} m, typically in Å range. Corresponds to frequencies (υ) of 3×10^{16}–3×10^{19} Hz, and energies (E) of ~0.124–1240 keV.



https://mynasadata.larc.nasa.gov/basic-page/electromagnetic-spectrum-diagram

$$E = h\upsilon$$

Energy = Planck's constant × Frequency
with $h = 6.626 \times 10^{-34}$ J.s

$$\lambda\upsilon = c$$

Wavelength × Frequency = speed of light
with $c = 3 \times 10^{8}$ m s^{-1}
And 1 eV = 1.6022×10^{-19} J

$$E = hc / \lambda \longrightarrow 12.4 \text{ keV @ } 1 \text{ Å}$$

Two types of X-rays

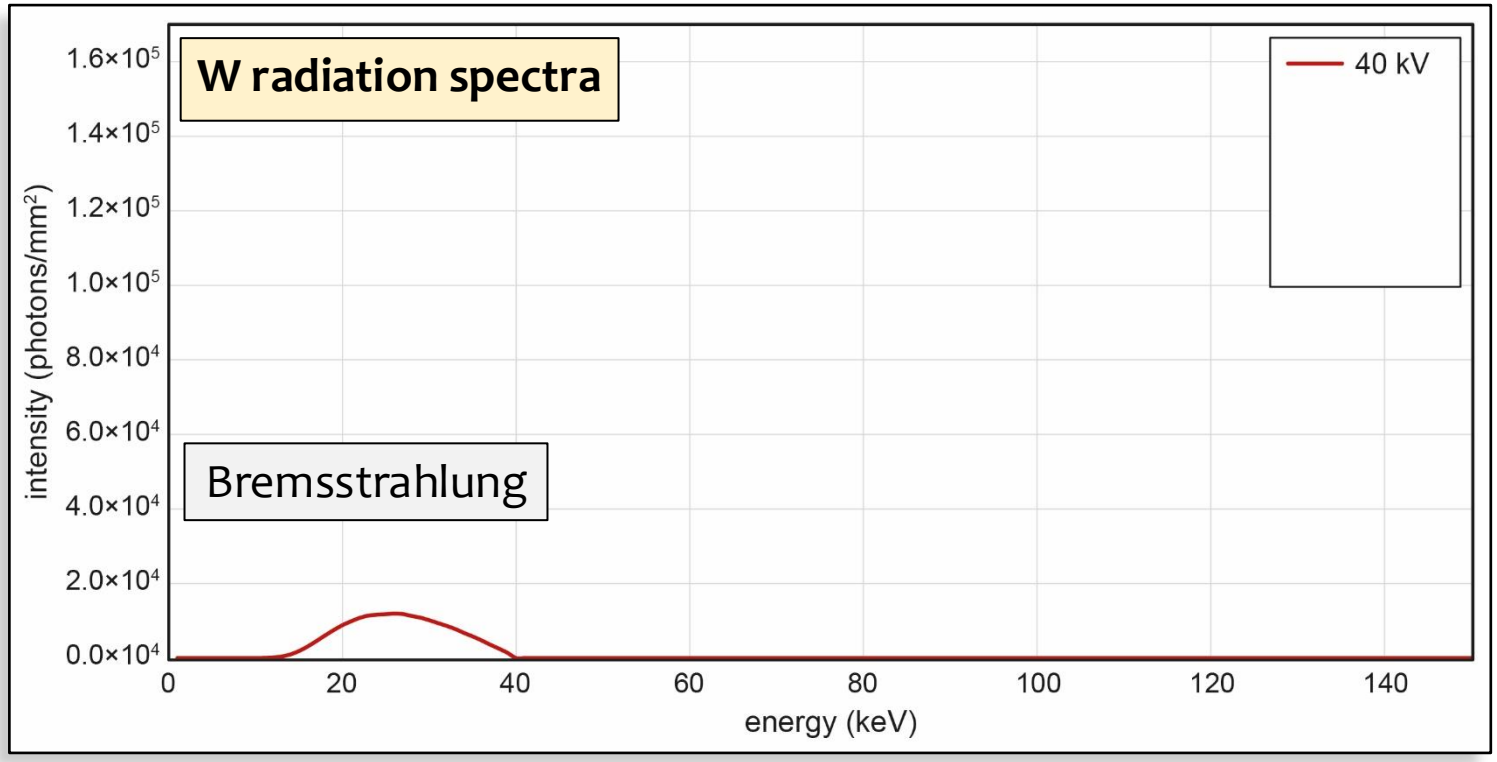
Production of X-rays involves bombardment of target with **energetic e^-**. These e^- undergo collisions and scattering during slowing down, resulting in production of **Bremsstrahlung** and **Characteristic Radiation**.

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Bremsstrahlung (continuous X-ray): result from e^- changing direction when travelling in proximity to target nucleus.

Energy of emitted X-ray photon is subtracted from kinetic energy of incoming e^-.

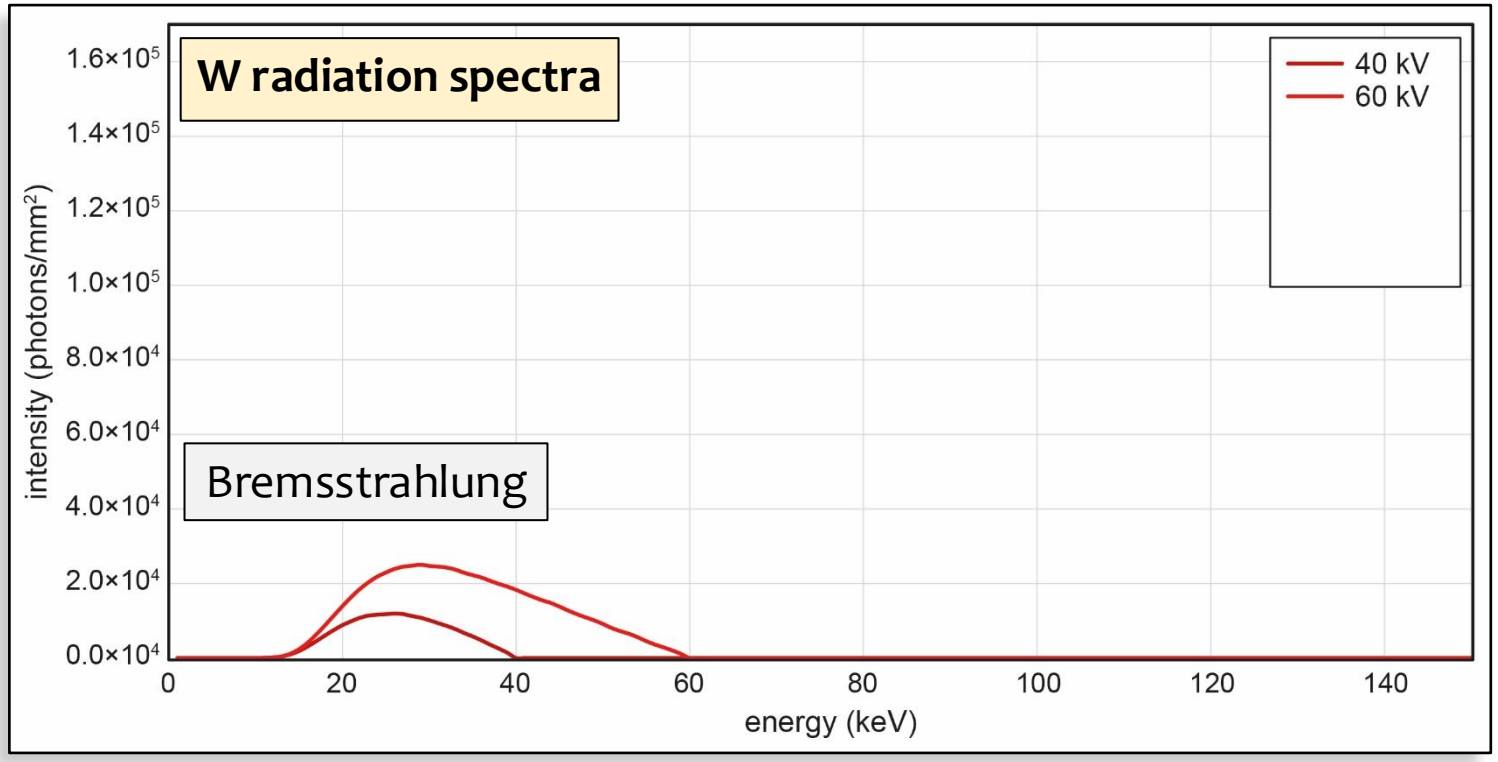


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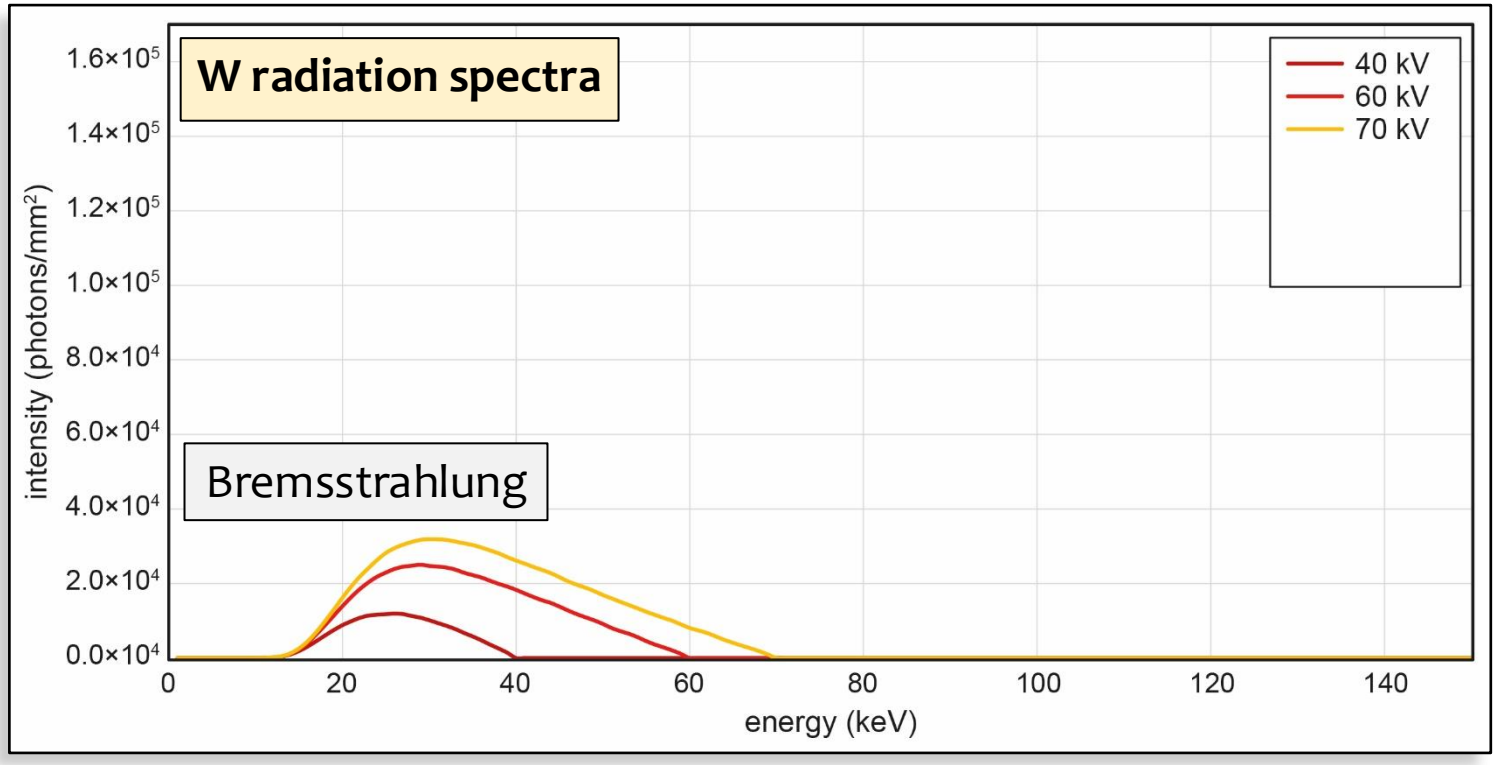


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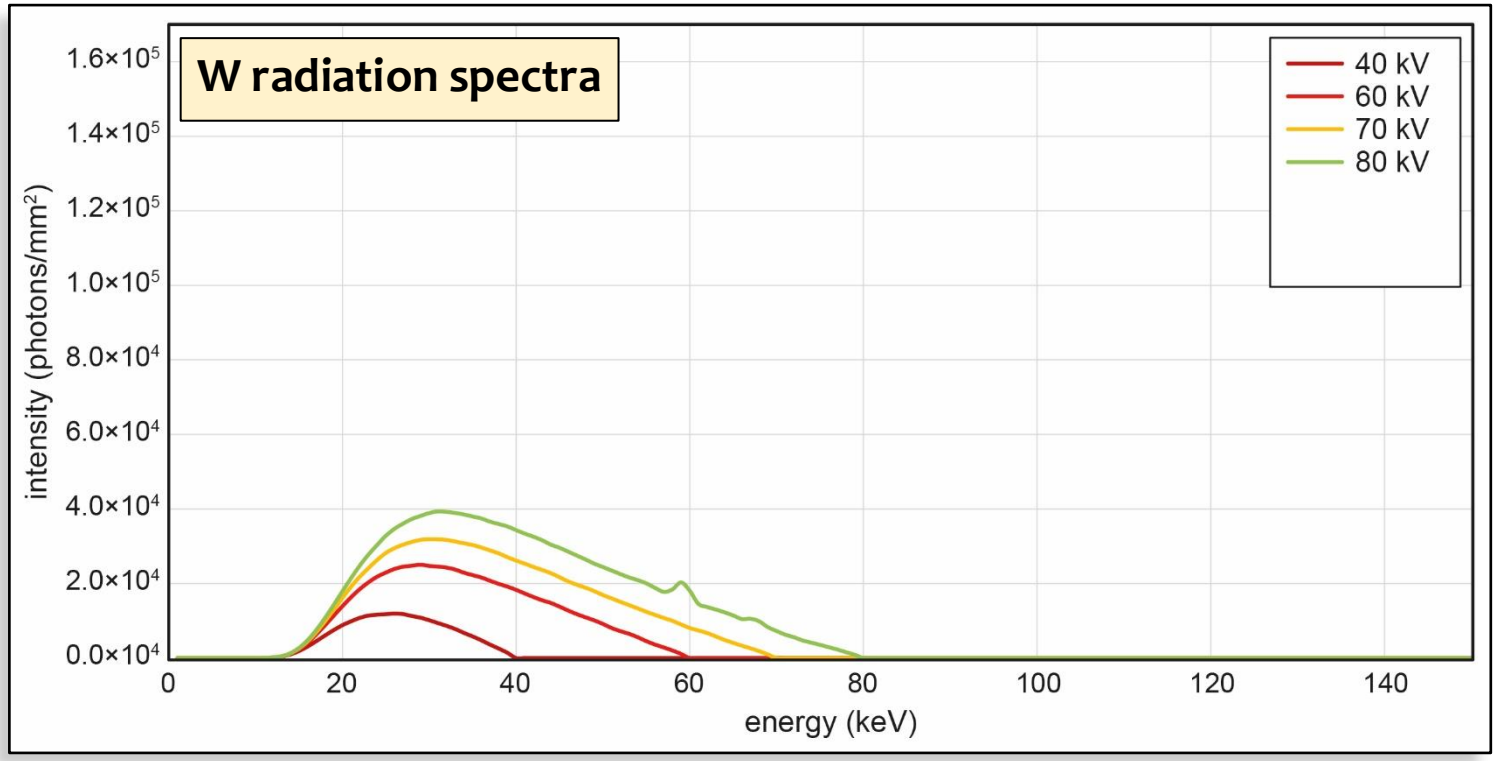


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Vacancy filled with e^- from outer shell → emission of X-ray photon with characteristic energy, equals to difference in binding energies of shells involved.

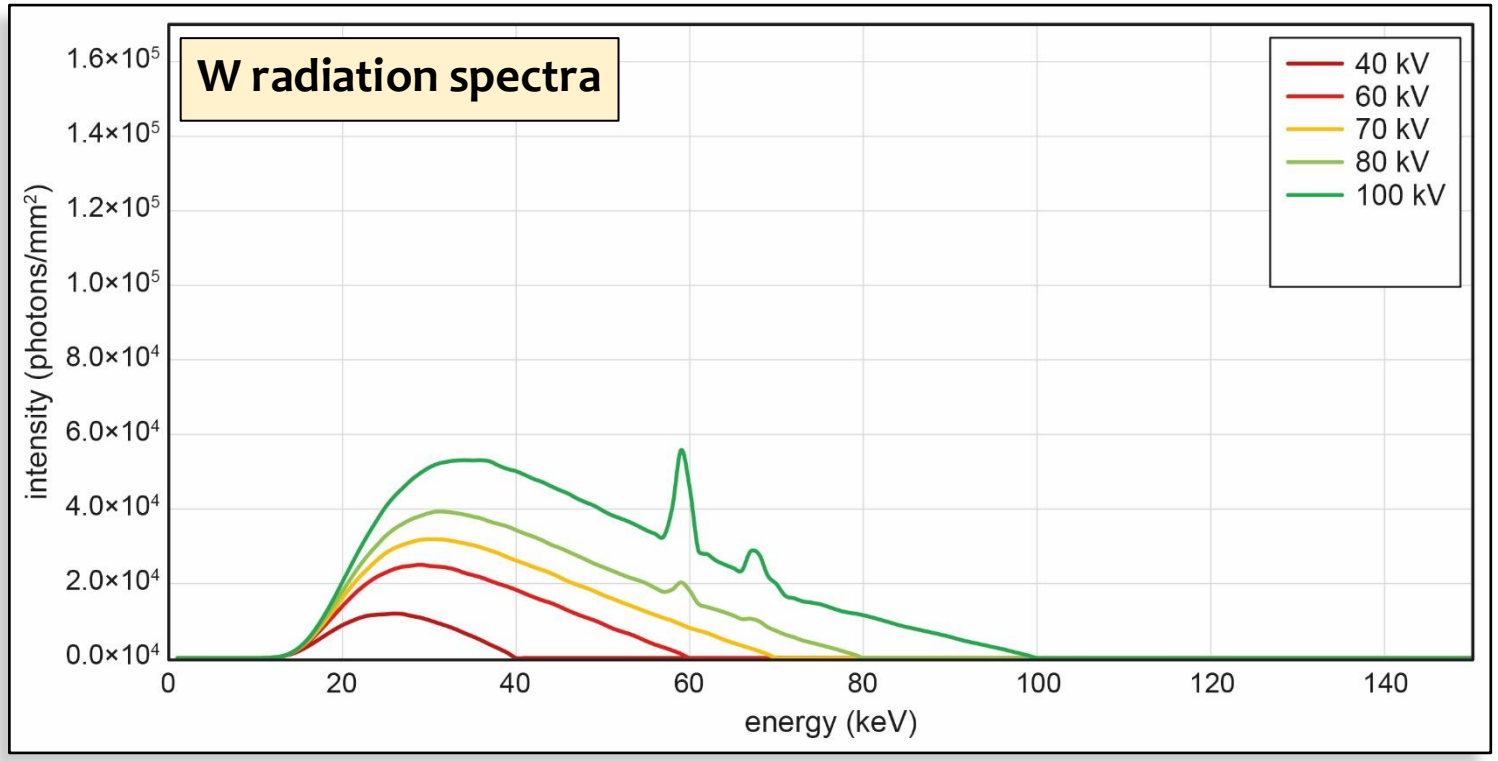


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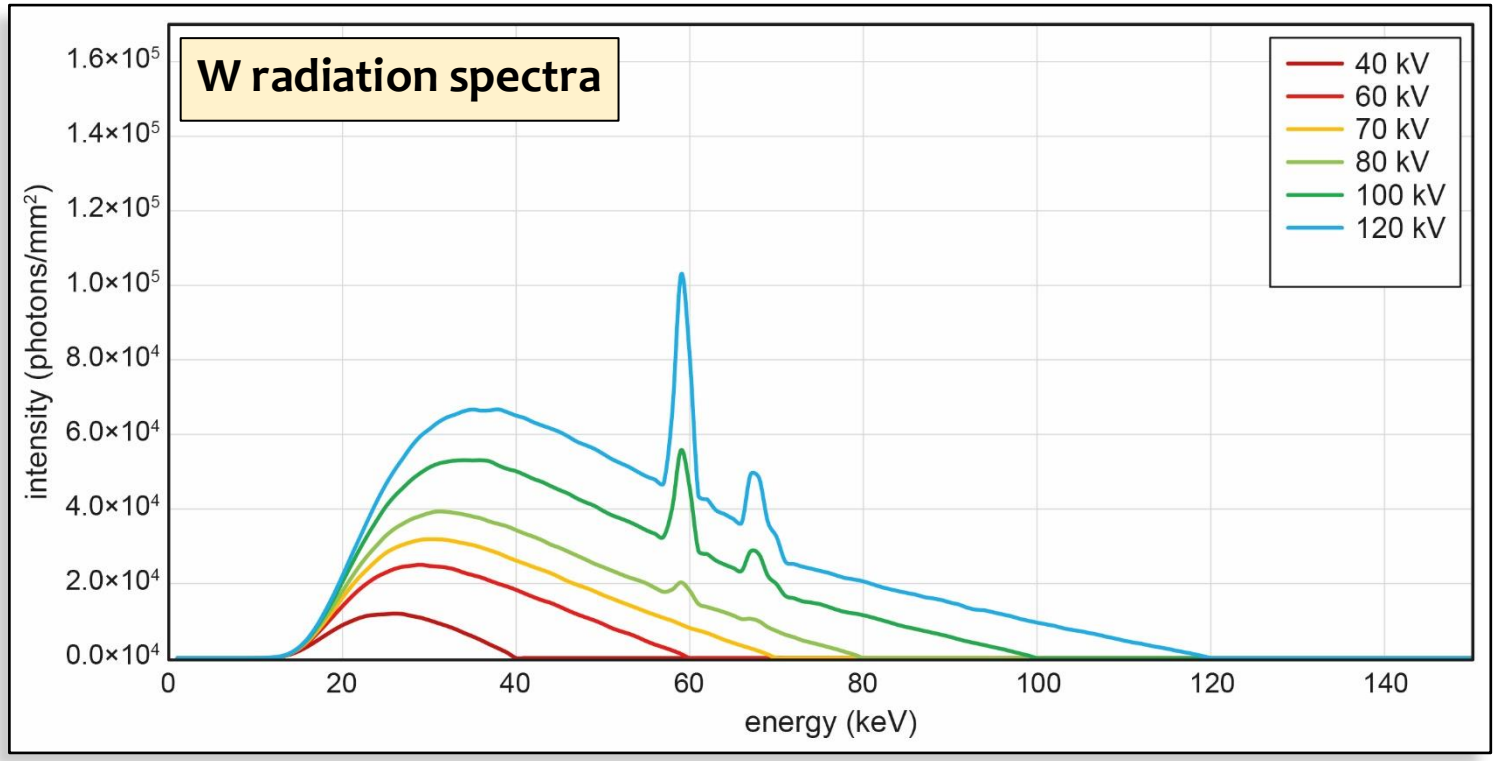


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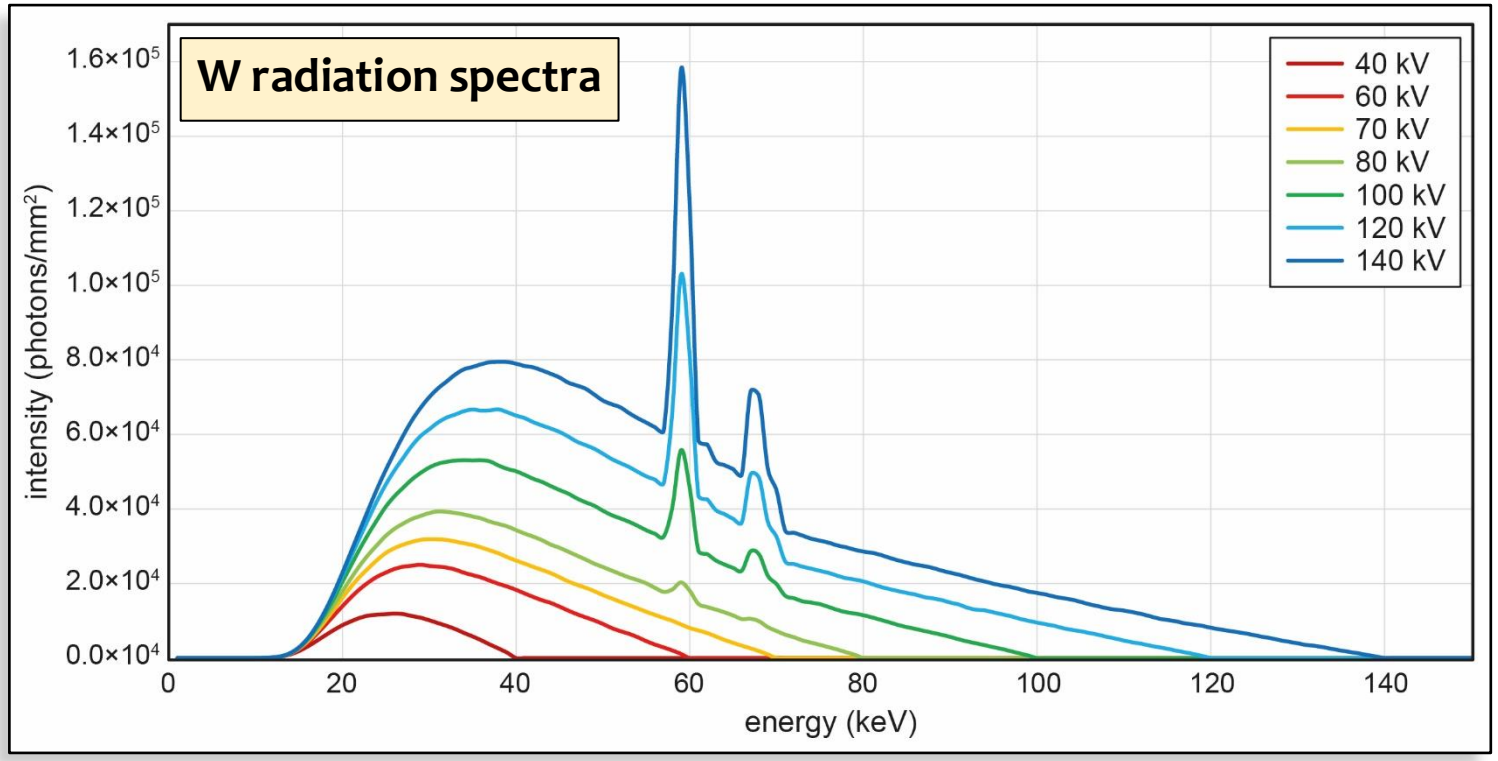


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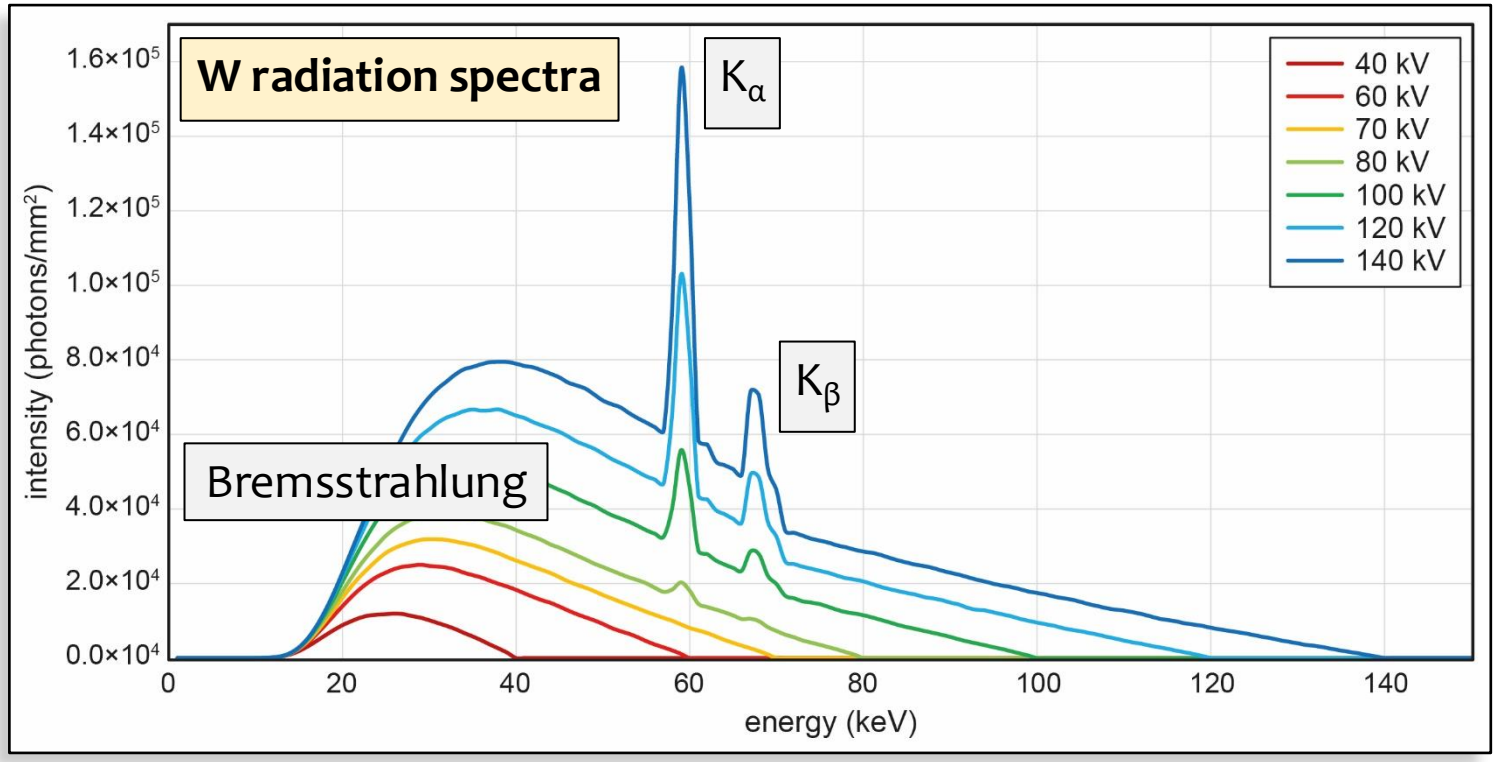


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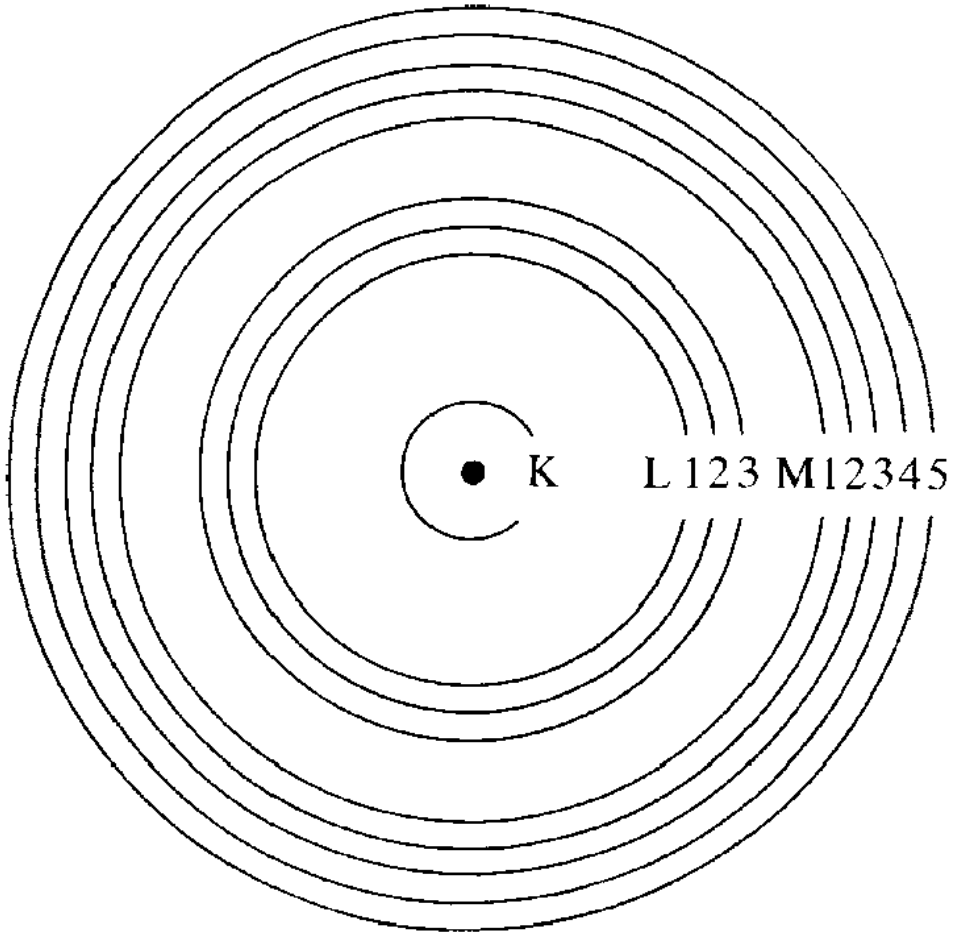
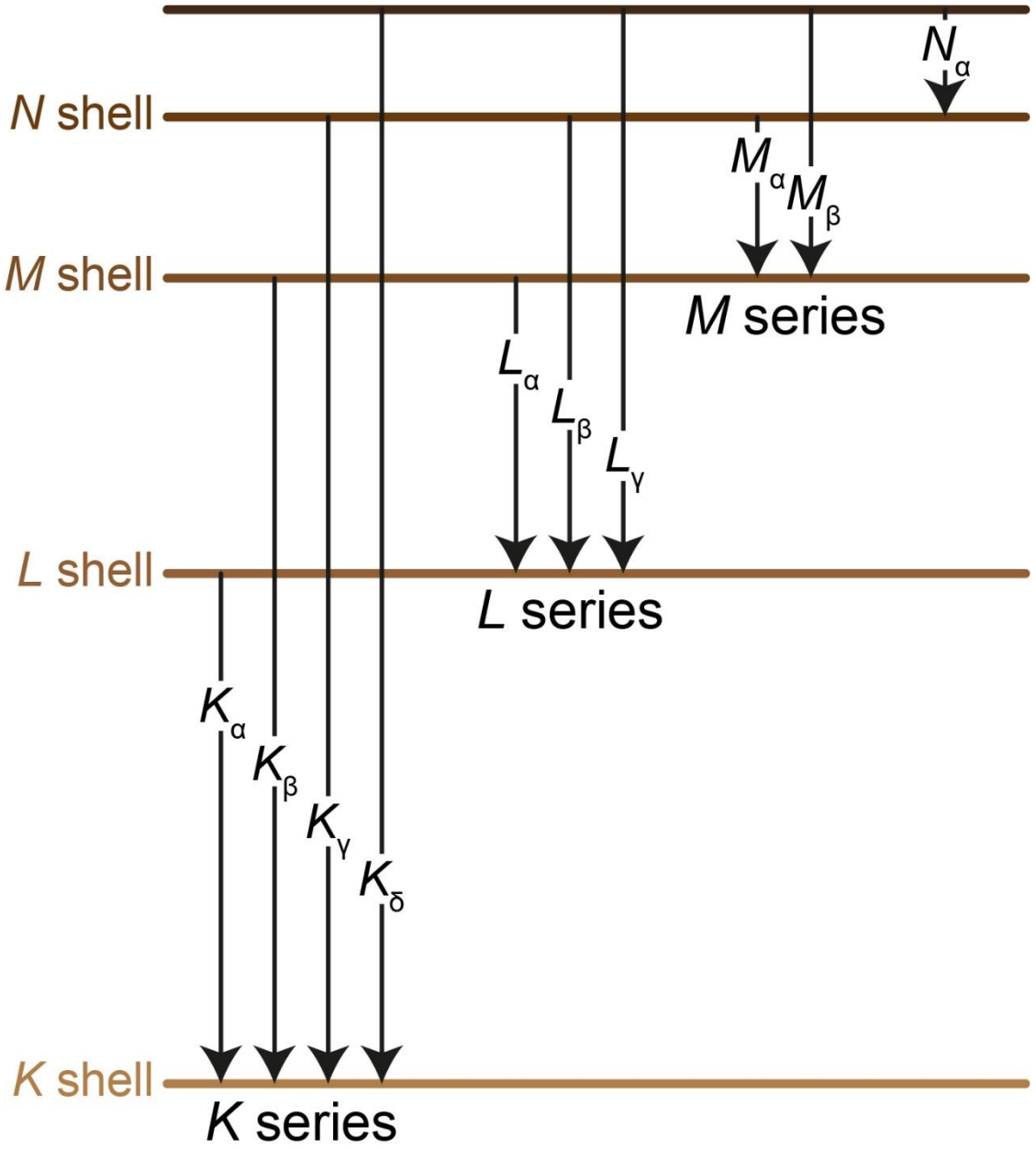


Fig. 2.7. A schematic diagram of inner atomic electron shells; characteristic X-rays are produced by transitions between these shells.

from Reed (2005) – *Electron Microprobe Analysis and Scanning Electron Microscopy in Geology*

Two types of X-rays – Characteristic



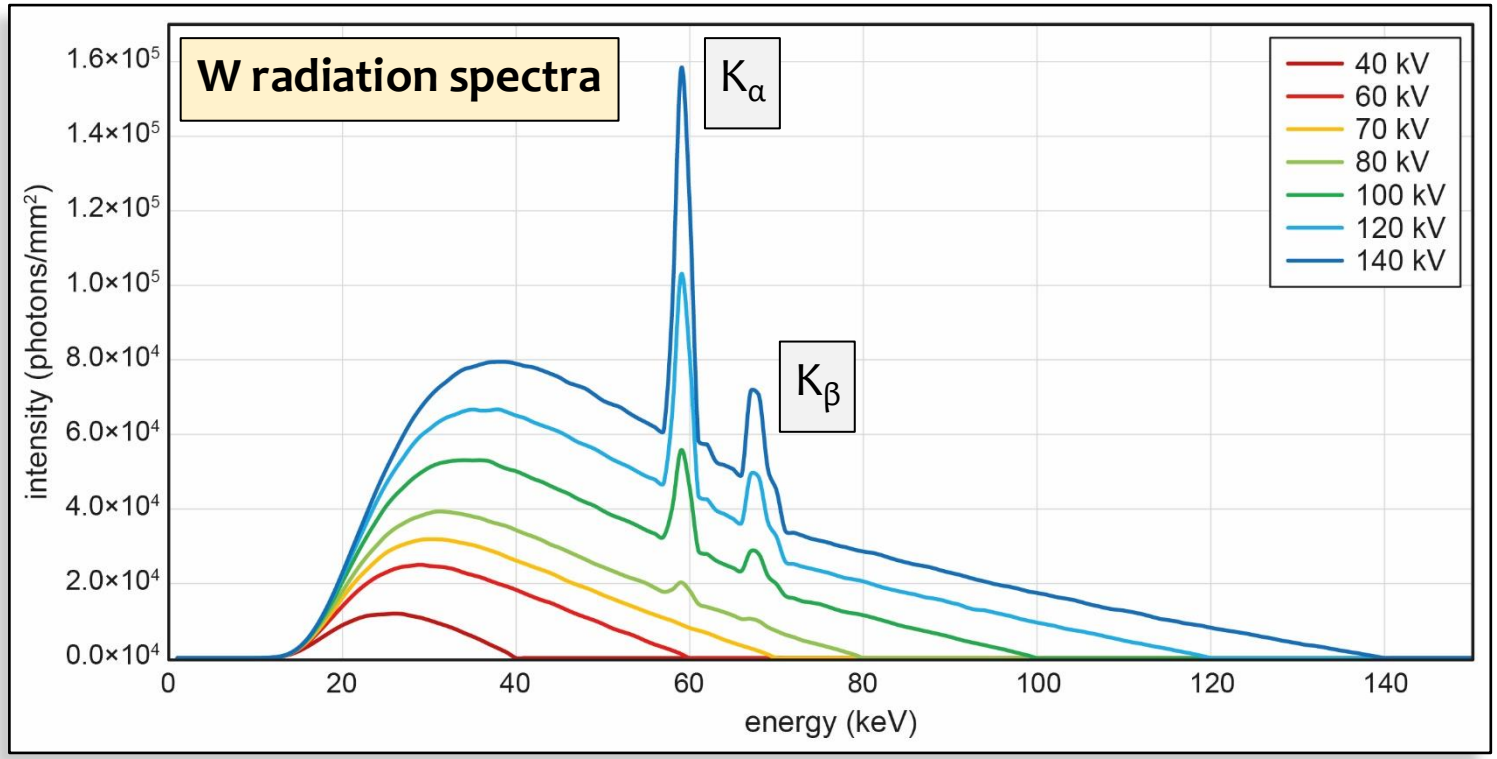
- K_α X-ray is produced due to removal of K shell electron, with L shell electron taking its place. K_β occurs when K shell electron is replaced by electron from the M shell.
- L_α X-ray is produced due to removal of L shell electron, replaced by M shell electron.
- M_α X-ray is produced due to removal of M shell electron, replaced by N shell electron.

Two types of X-rays – Characteristic

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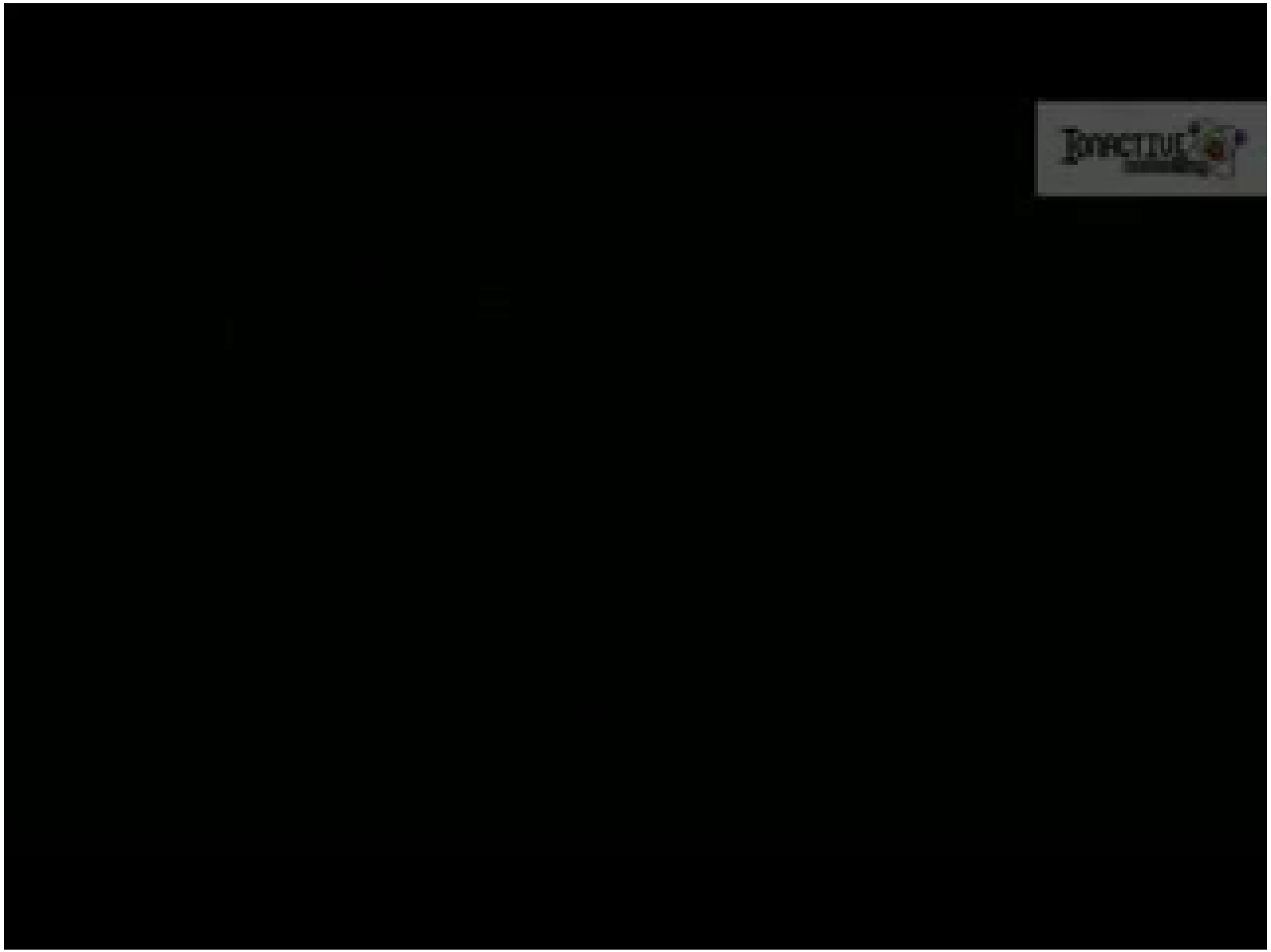
Characteristic X-ray:

Target	Z	K_α line
Cu	29	8.04 keV (λ=1.54 Å)
Mo	42	17.5 keV (λ=0.71 Å)
Ag	47	22.2 keV (λ=0.56 Å)
W	74	59.3 keV (λ=0.21 Å)



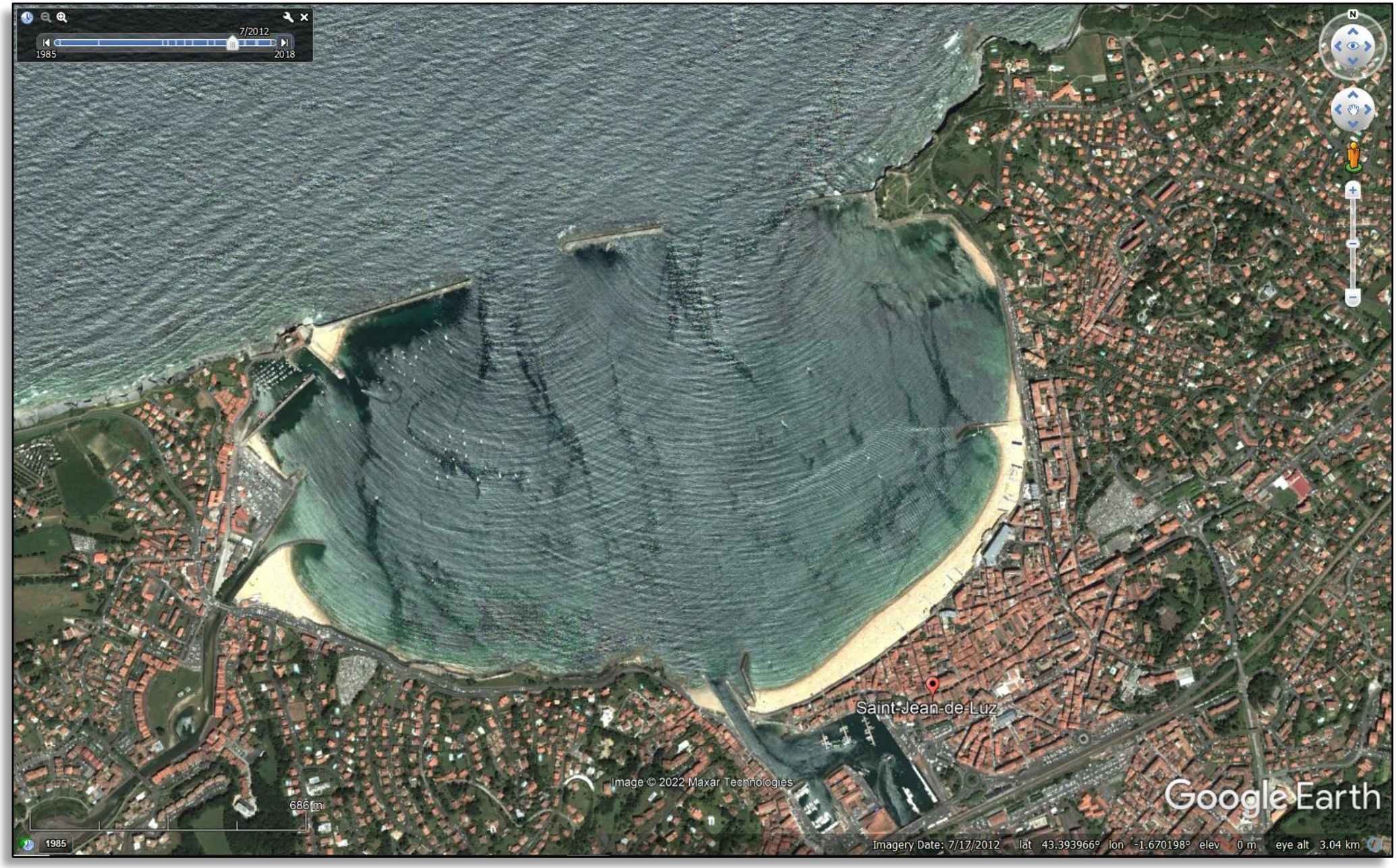
→ characteristics X-ray are specific to each elements.

Two types of X-rays



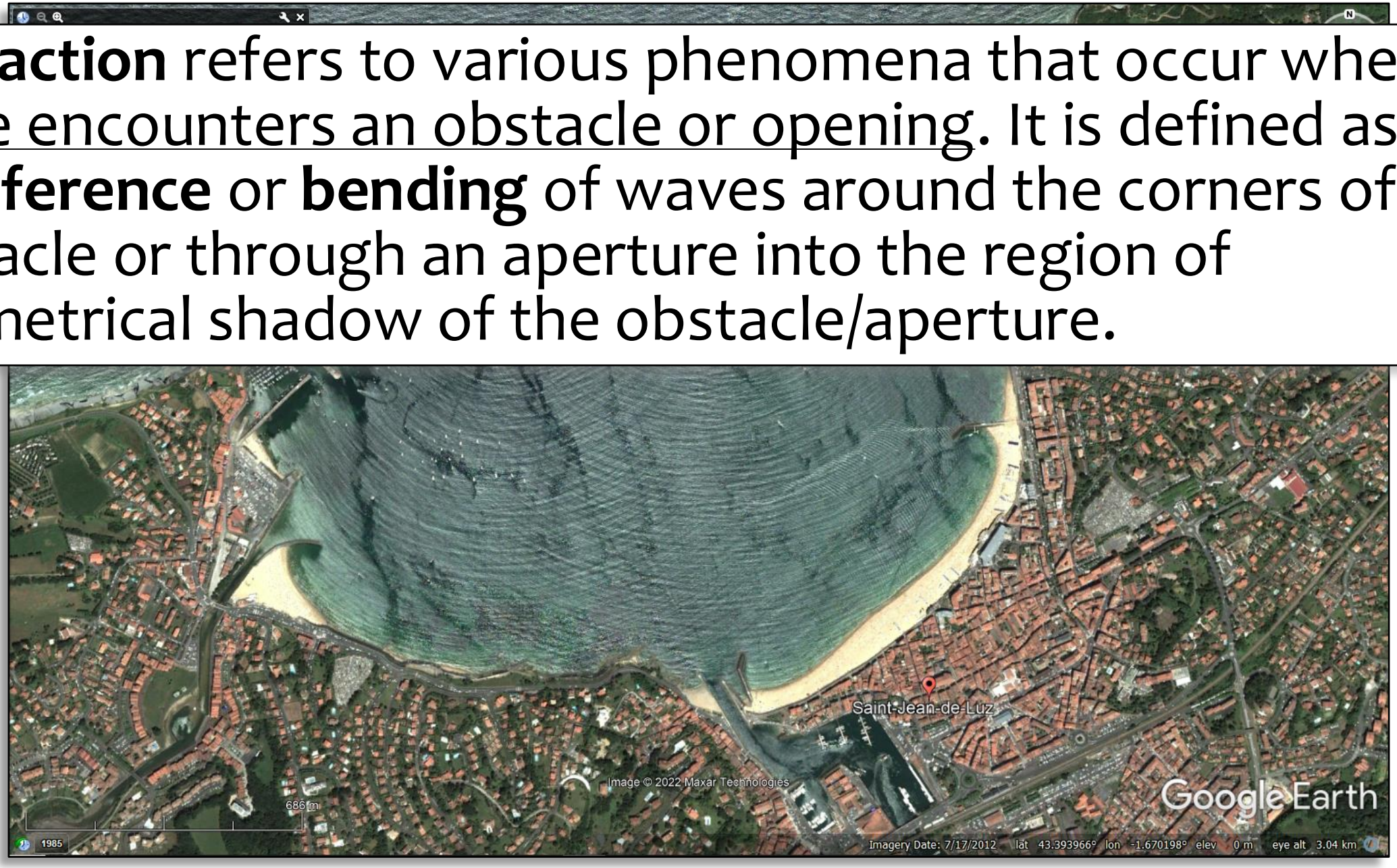
https://www.youtube.com/watch?v=BcoeOjWkxpU&list=PLRTZrq1YpLbhPIV4c7Cg95GAAJ1i3pAlu&index=2

What is diffraction?



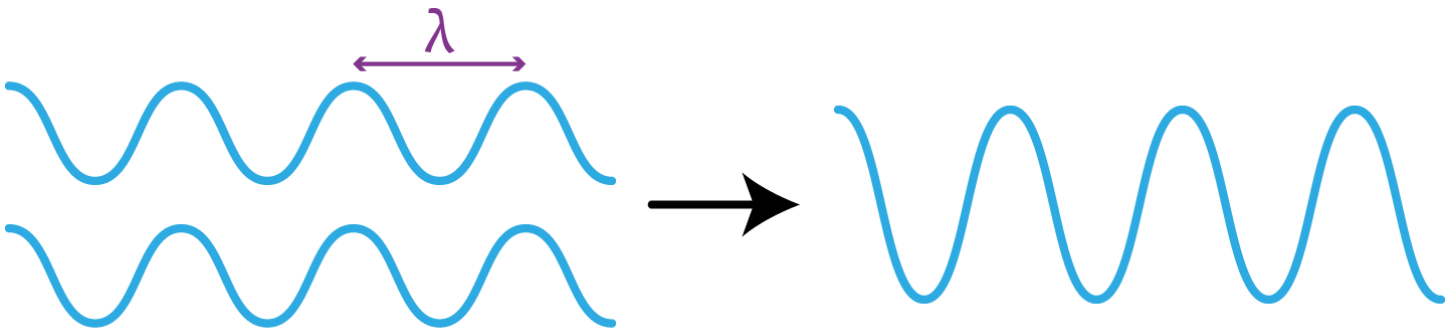
What is diffraction?

Diffraction refers to various phenomena that occur when a wave encounters an obstacle or opening. It is defined as the **interference** or **bending** of waves around the corners of an obstacle or through an aperture into the region of geometrical shadow of the obstacle/aperture.



Interference and diffraction

When 2 sets of waves meet, they combine to produce a new pattern.

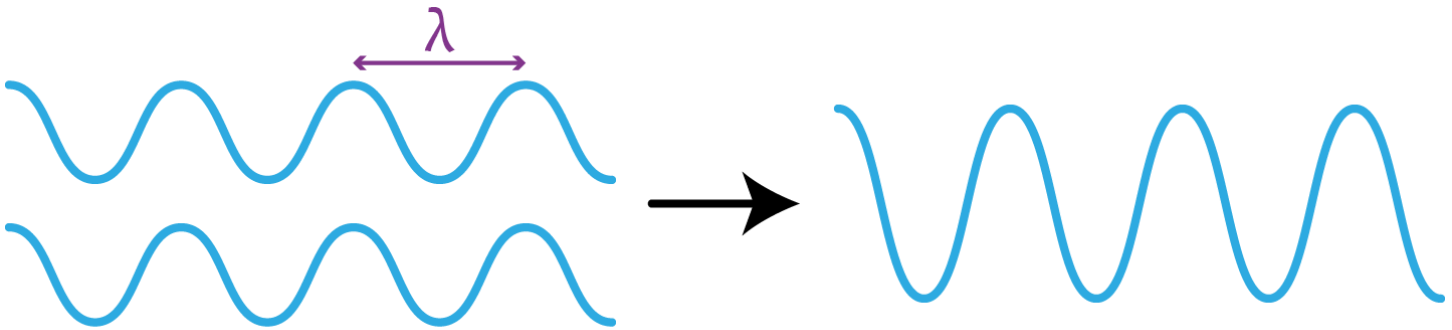


<u>Constructive interference</u>: 2 sets of waves meet <u>in phase.</u>

Occurs when path-difference between 2 waves is $n\lambda$, with n an integer.

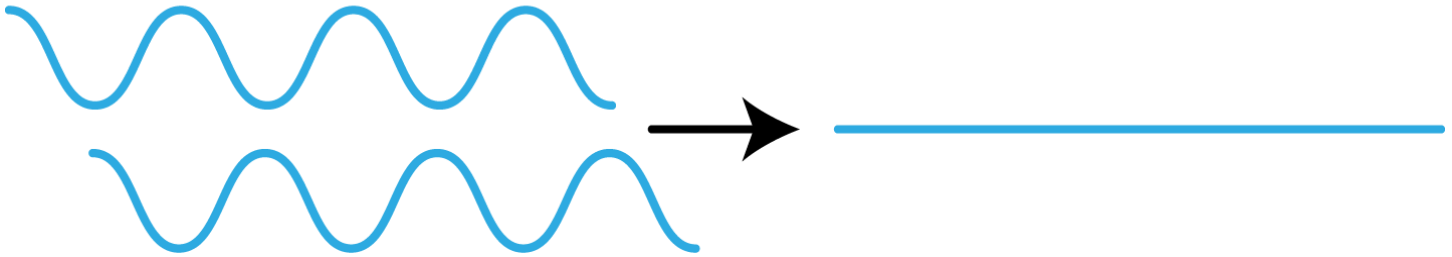
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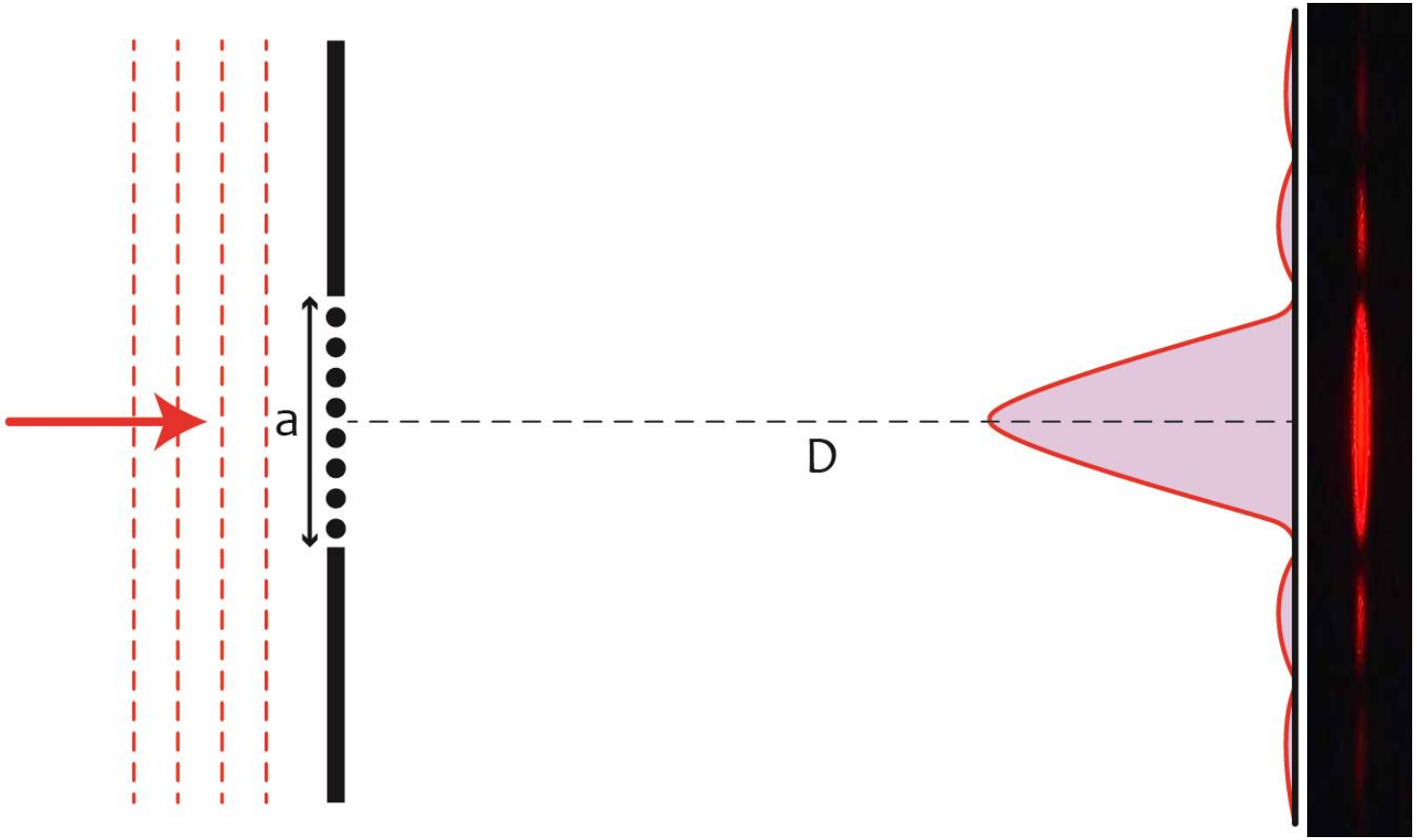
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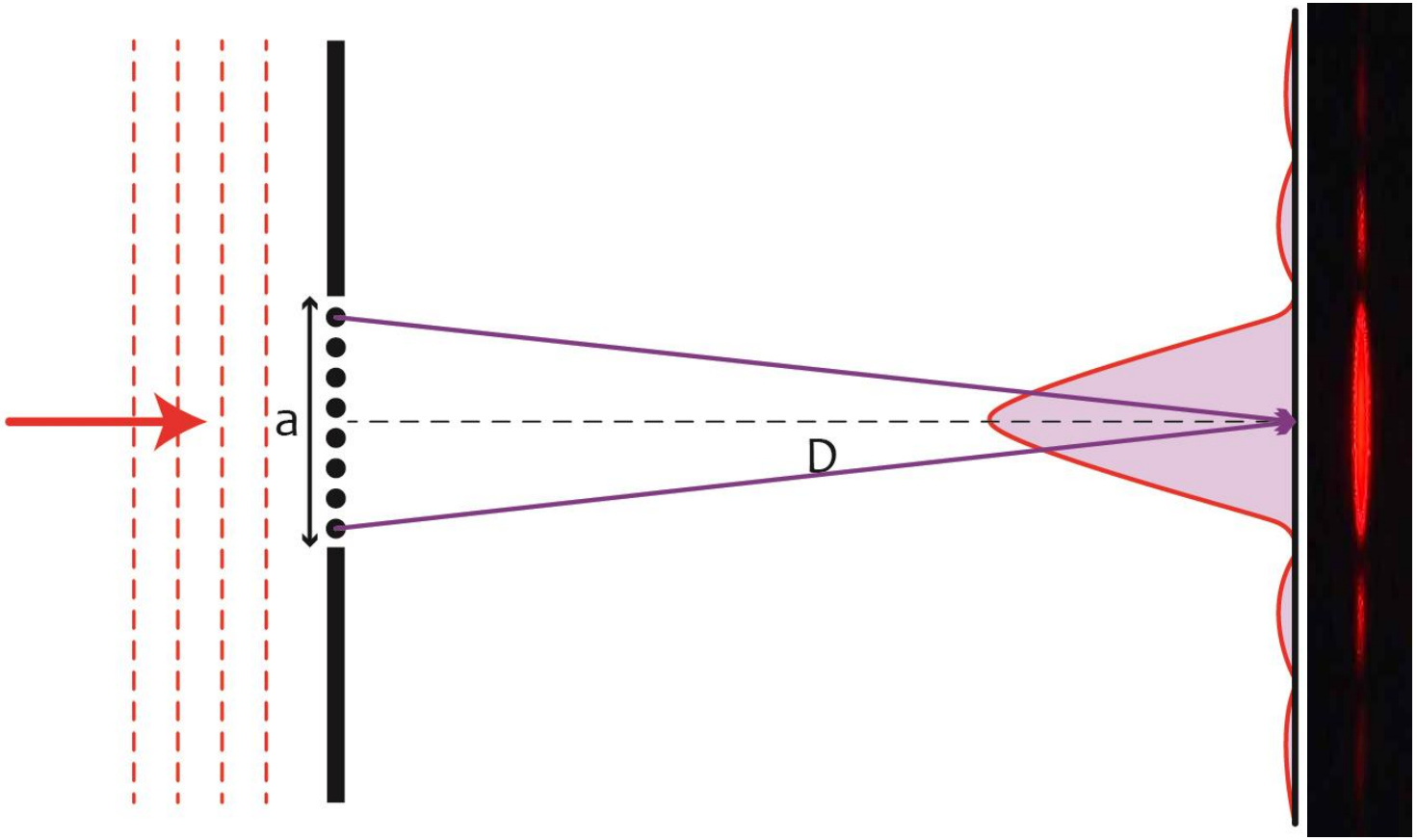
<u>Destructive interference</u>: 2 sets of waves meet completely <u>out of phase.</u>

Occurs when path-difference between 2 waves is $(n+1/2)\lambda$, with n an integer.

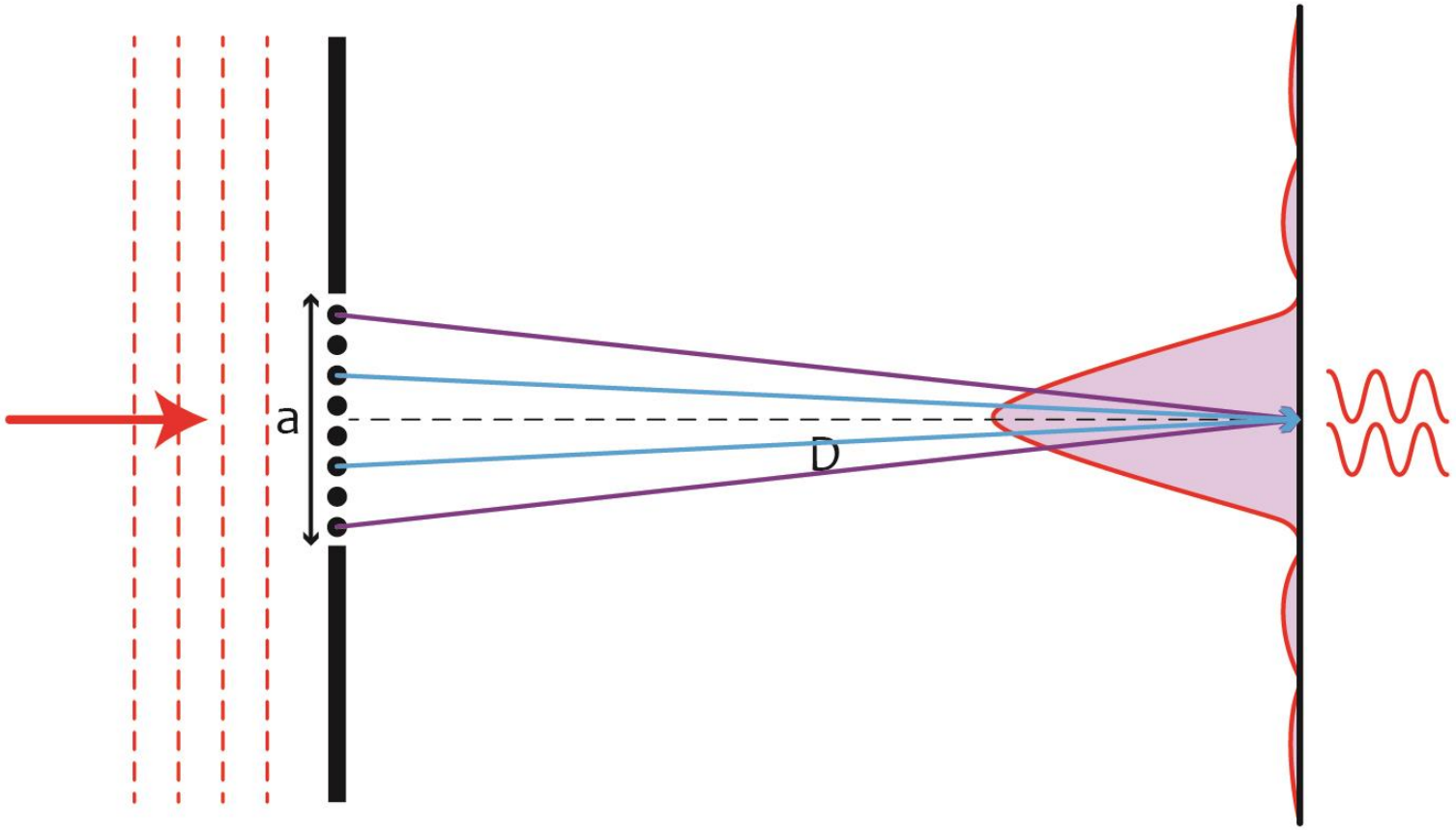
Fraunhofer single slit diffraction



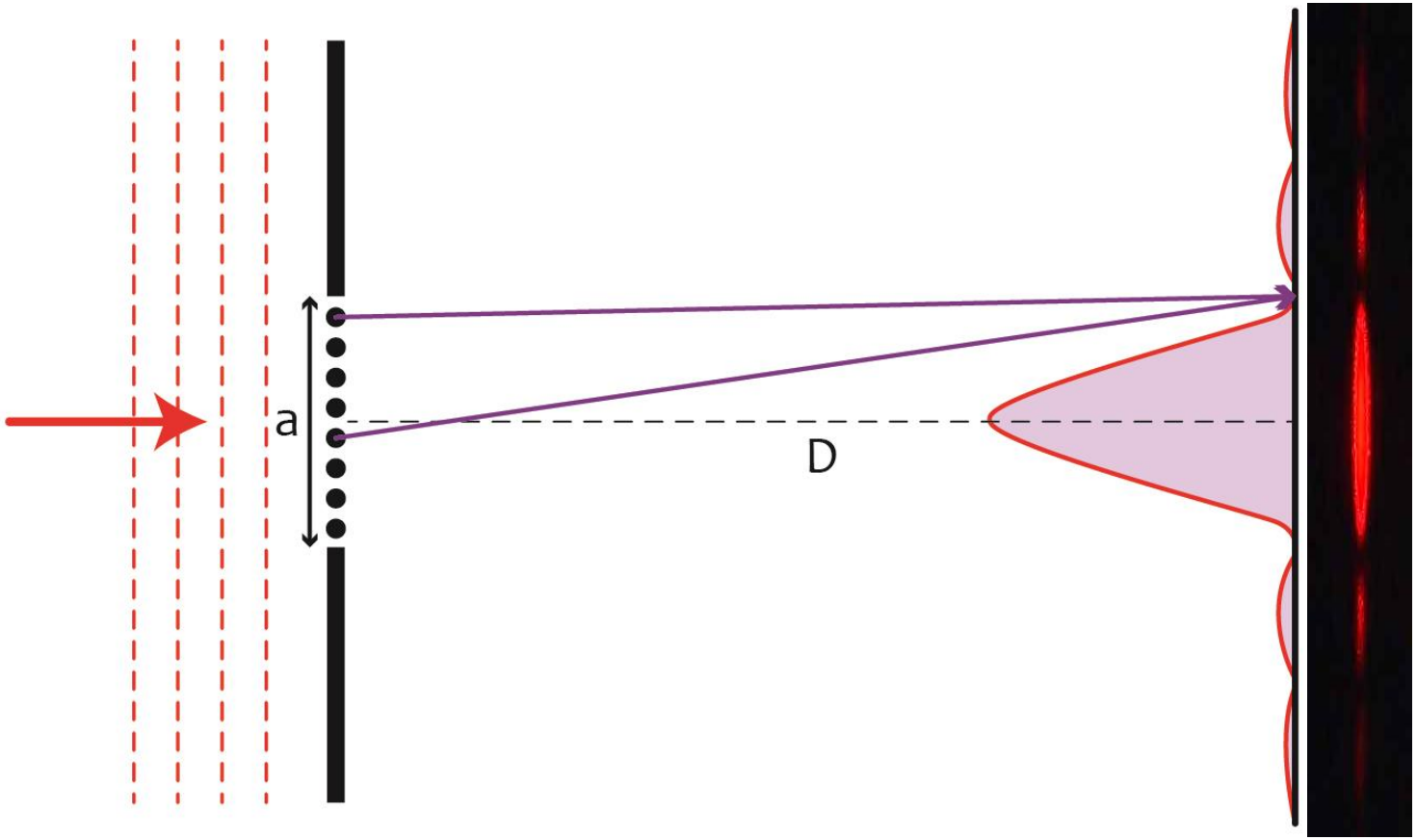
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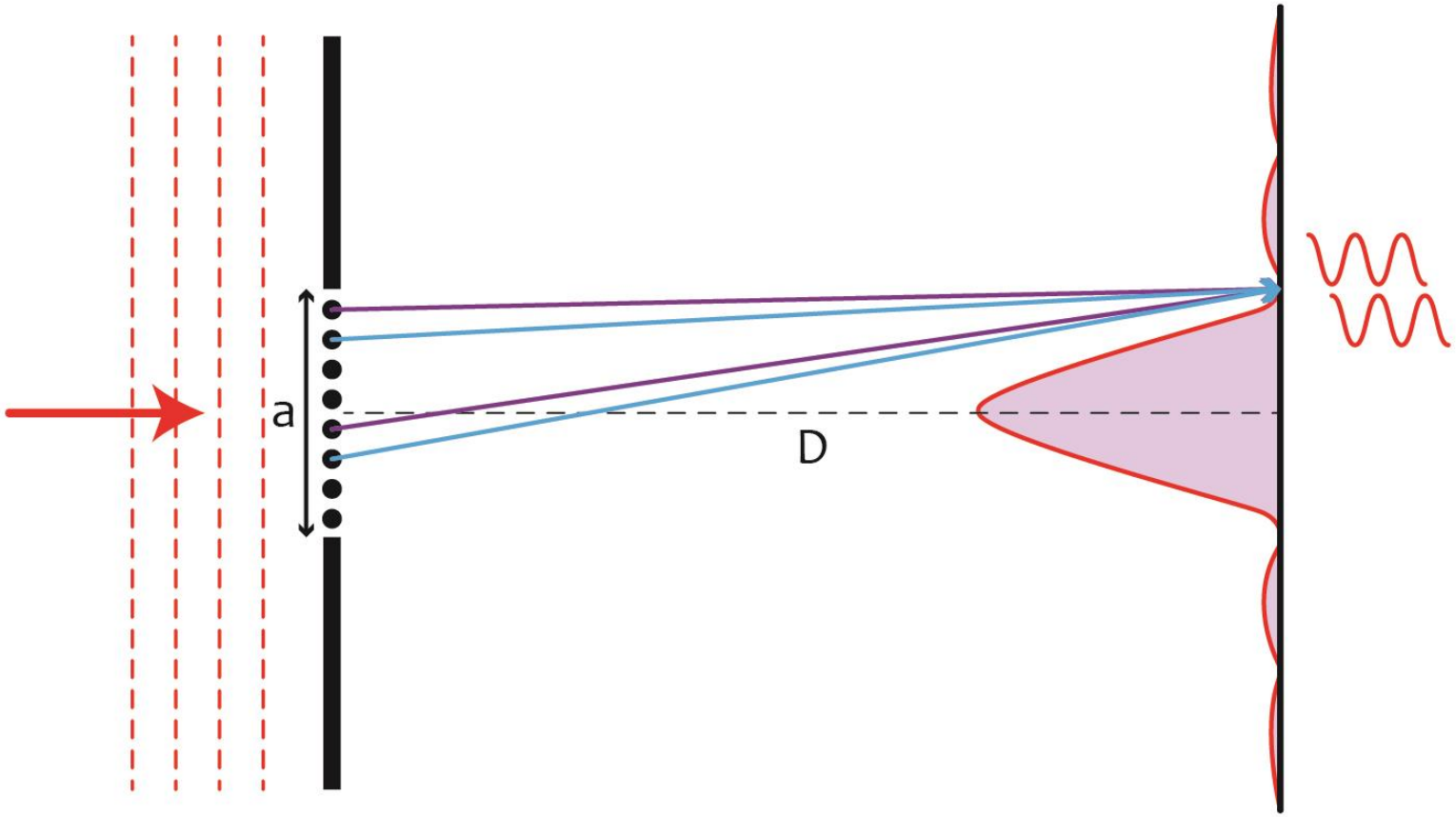
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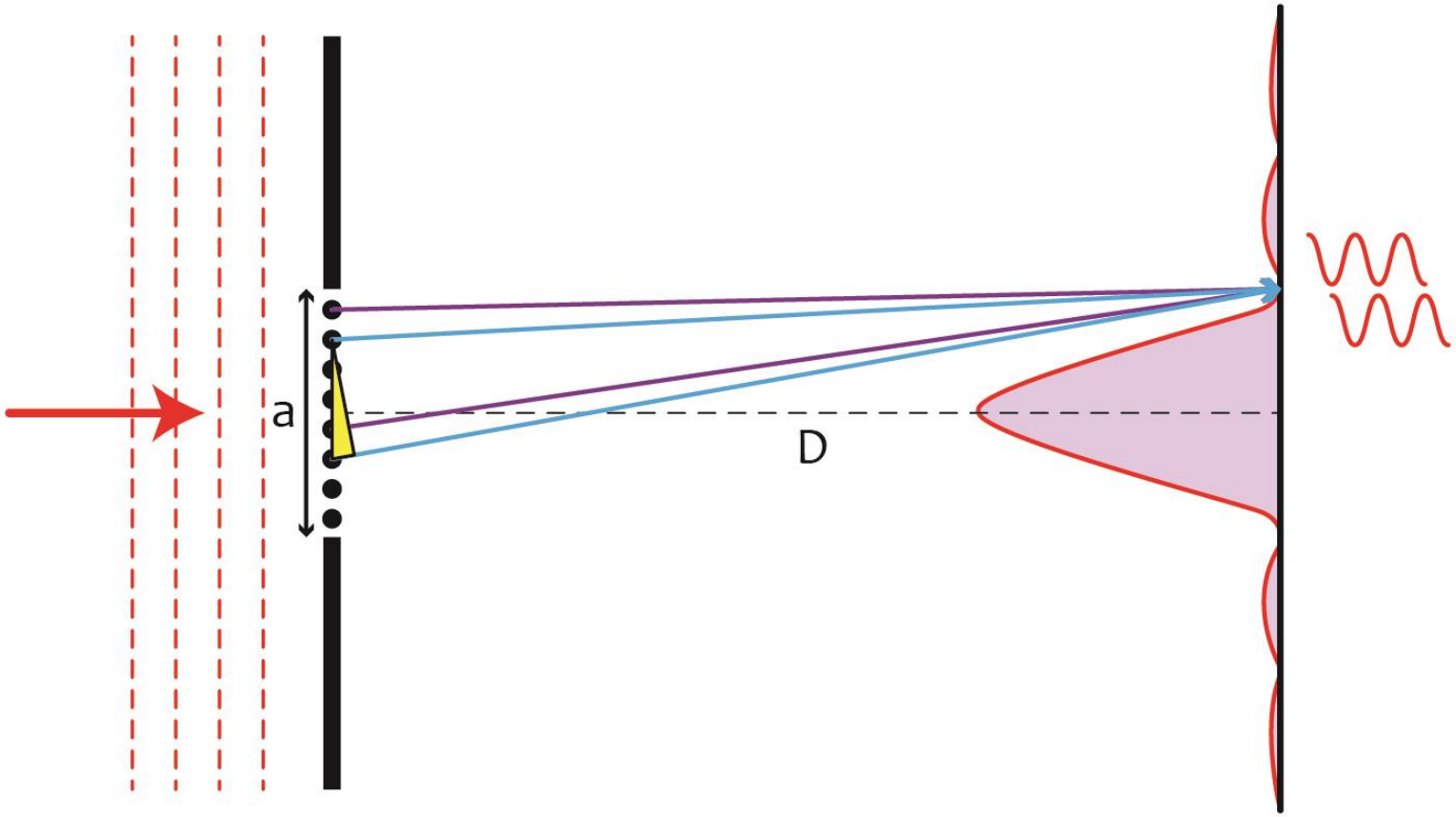
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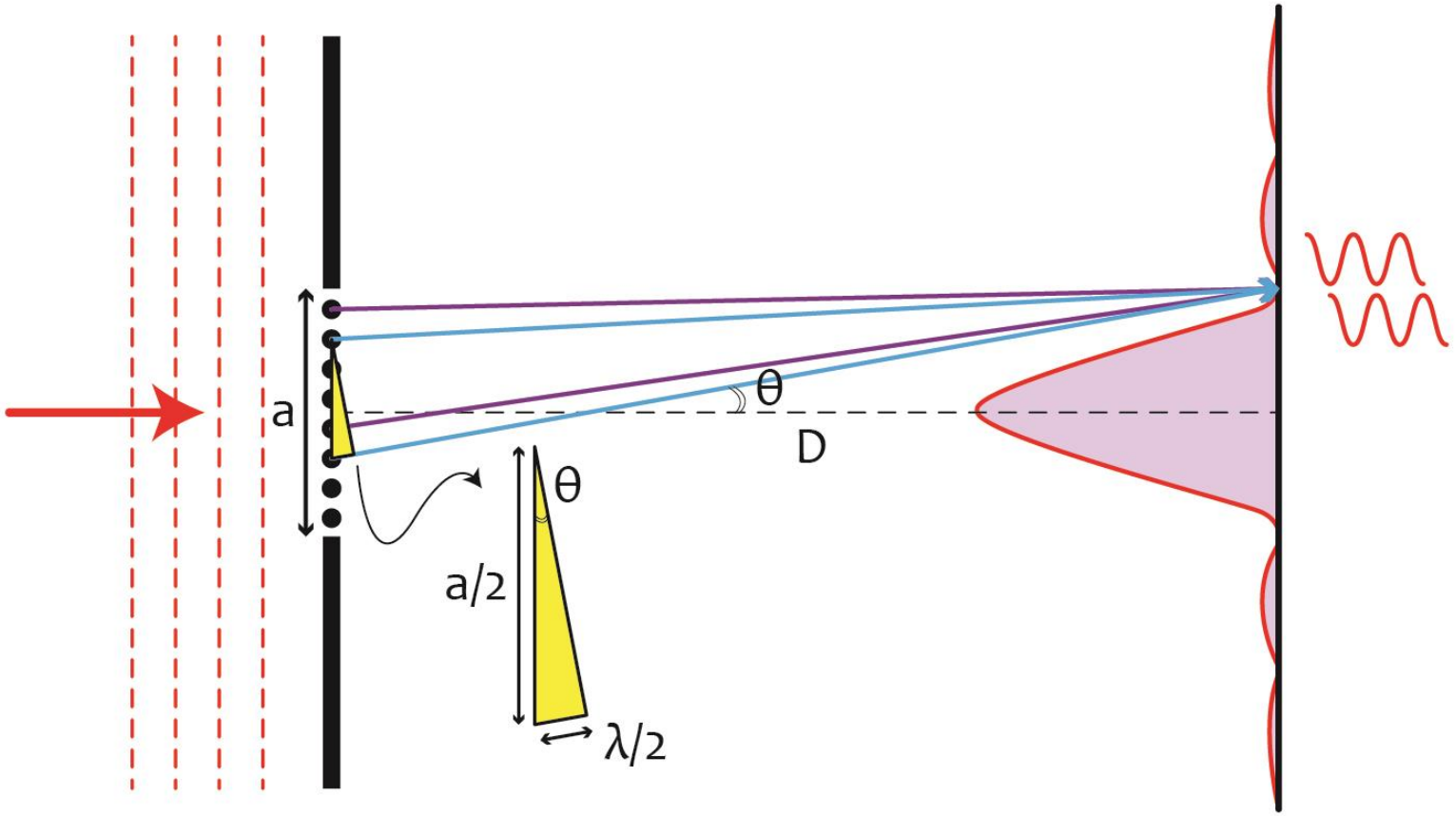
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Fraunhofer single slit diffraction



<u>Condition for minimum</u>:

Sine = opposite/hypotenuse

→ $\sin(\theta) = (\lambda/2) / (a/2)$
→ $(a/2).\sin(\theta) = (\lambda/2)$
→ $a.\sin(\theta) = \lambda$

Generalised for any integer *m*

→ $\mathbf{a.\sin(\theta) = m.\lambda}$

➔ Occurs because of **<u>path length difference</u>**.

Diffraction grating

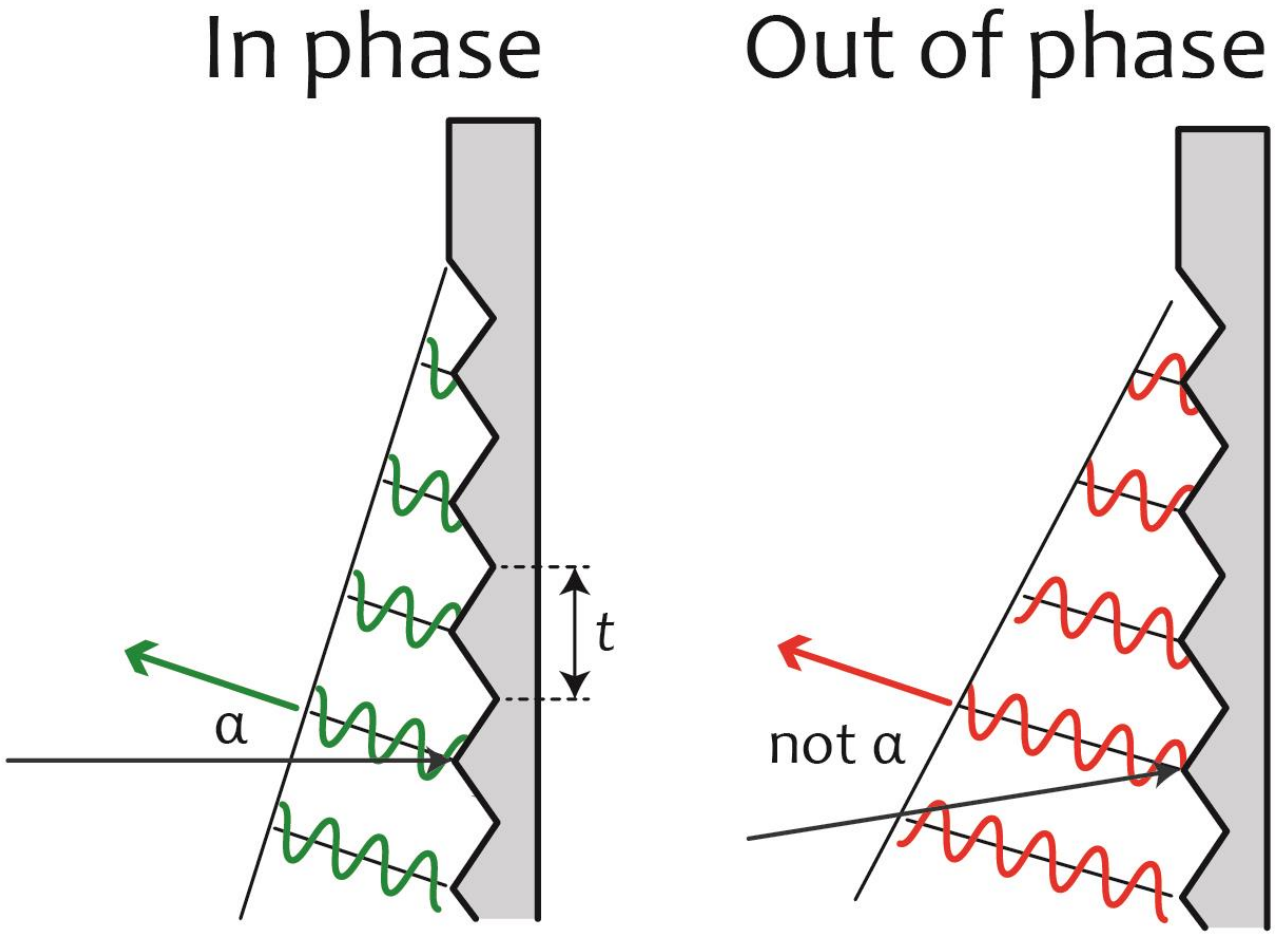
Here again, it's all to do with **path length difference**.

When path difference between 2 waves is multiple of λ, waves in phase → maximum intensity.

For **monochromatic light**:

$$n.\lambda = t.\sin(\alpha)$$

For example, for red laser light with λ = 650 nm and a grating with t = 1 μm, max. intensity at α = 40.5°.



Diffraction grating

Here again, it's all to do with **path length difference**.

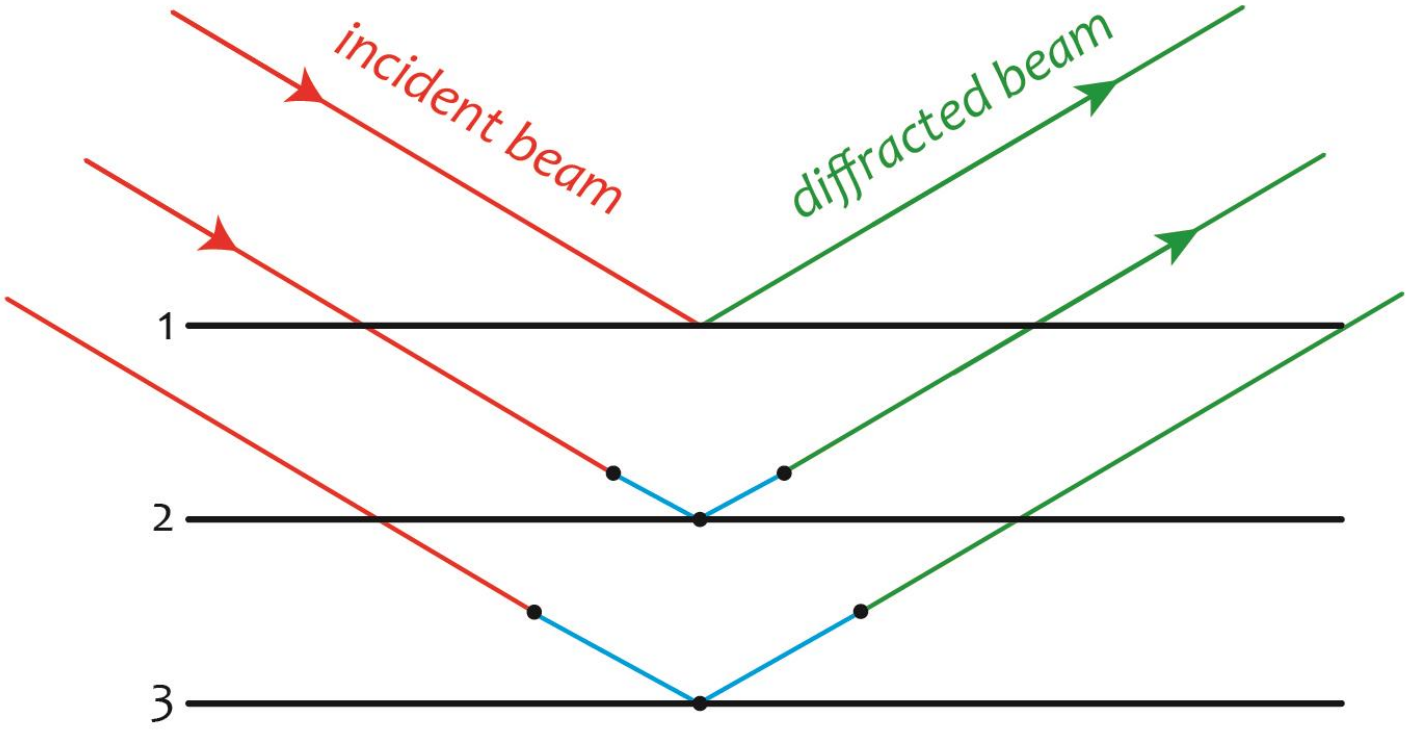
When path difference between 2 waves is multiple of λ, waves in phase → maximum intensity.

Also works with white light, and this will be relevant to ICP-OES, which we will talk about next week!



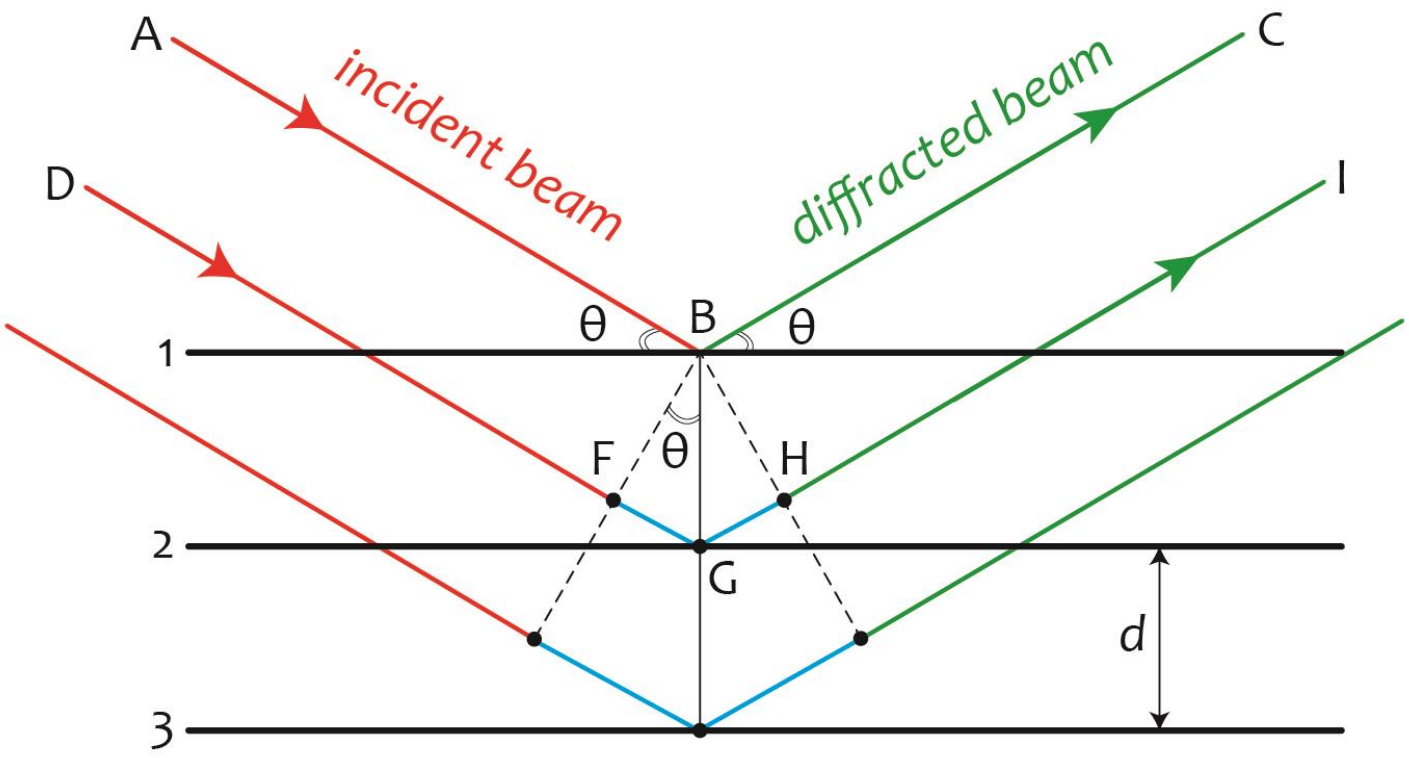
Diffraction from crystal lattice planes

Distances between crystal lattice planes ~ X-rays wavelength ~ Angstrom.



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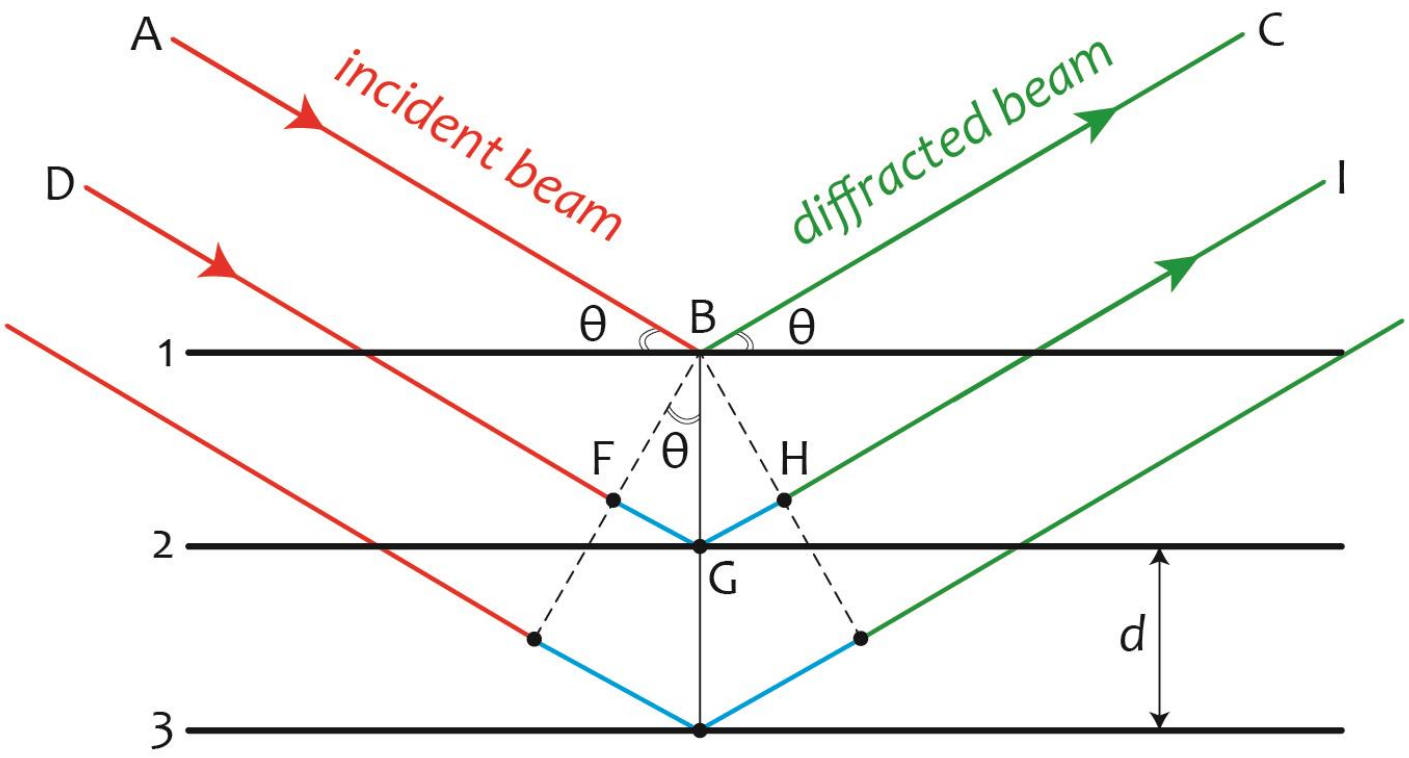
$\longrightarrow \sin(\theta) = FG/d = GH/d$

$\longrightarrow d.\sin(\theta) = FG = GH$

$\longrightarrow 2d.\sin(\theta) = FG + GH$

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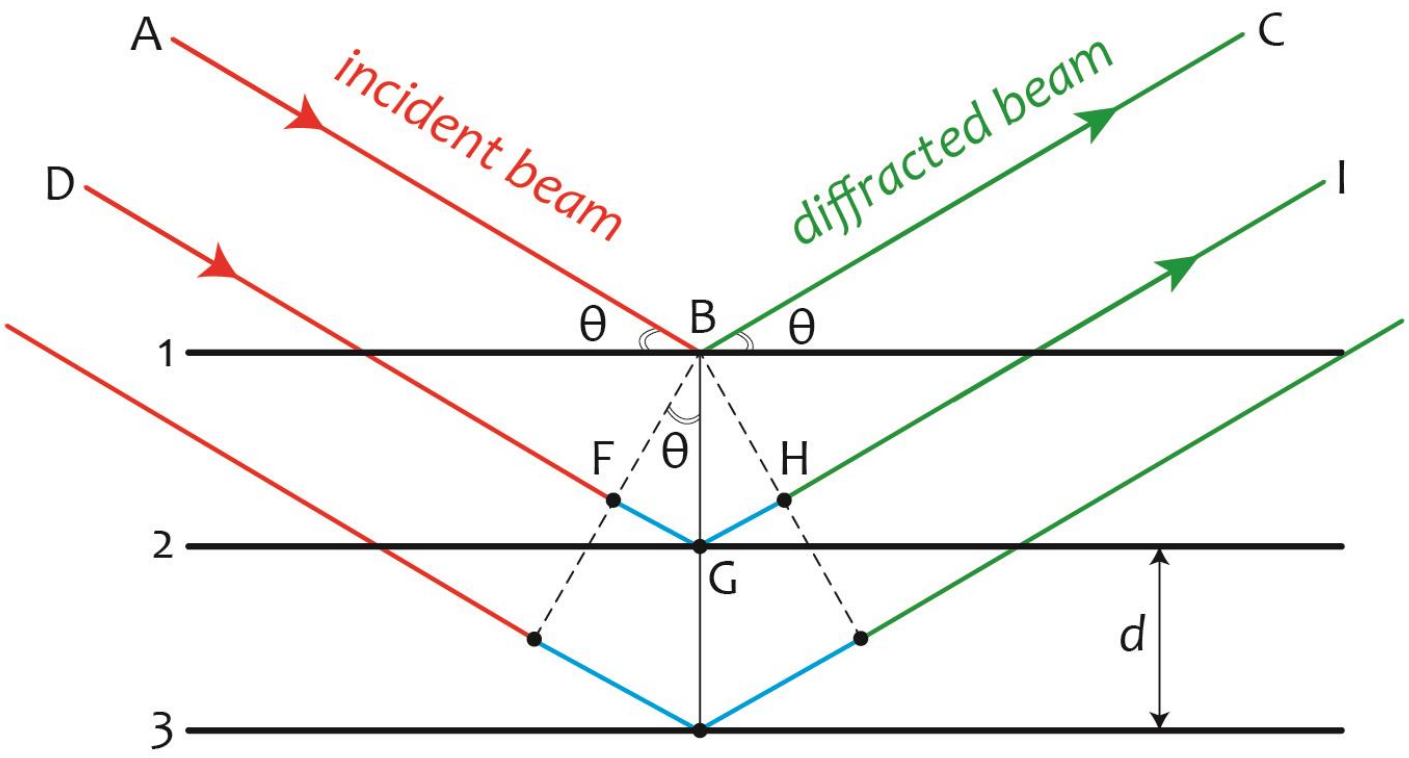
$\longrightarrow 2d.\sin(\theta) = FG + GH$

For constructive interference, total **path length change** between ABC & DGI (**FG+GH**) must be integer multiple of λ, or $n\lambda$.

So when **$FG+GH = \lambda = 2d.\sin(\theta)$** $\longrightarrow$ 1st order **constructive interference**

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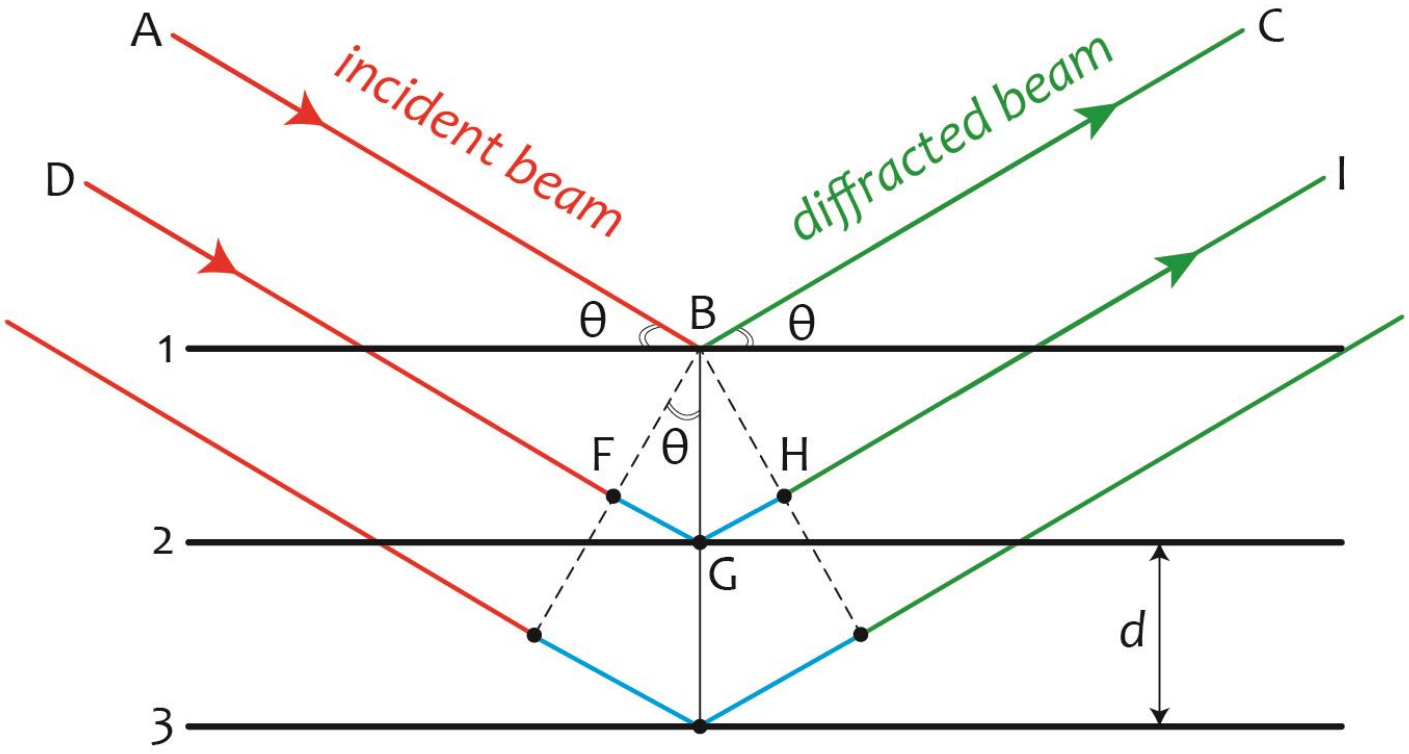
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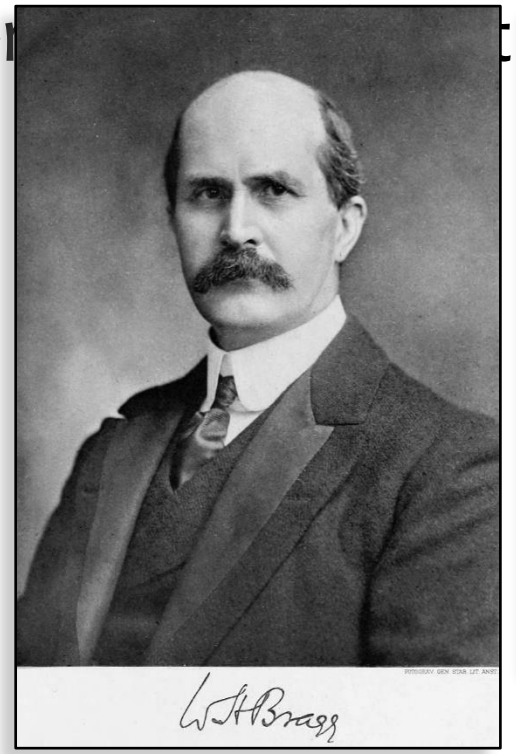
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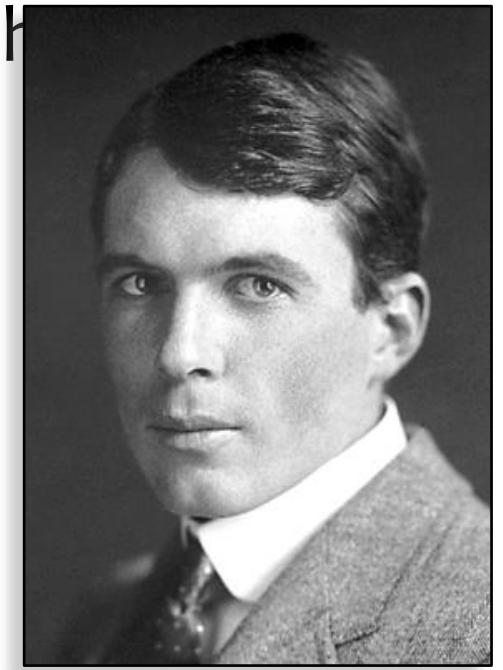
Generalised to all orders: $n\lambda = 2d.\sin(\theta)$ → Braggs' law

Diffraction from crystal lattice planes

Distances between crystal lattice planes ~ X-





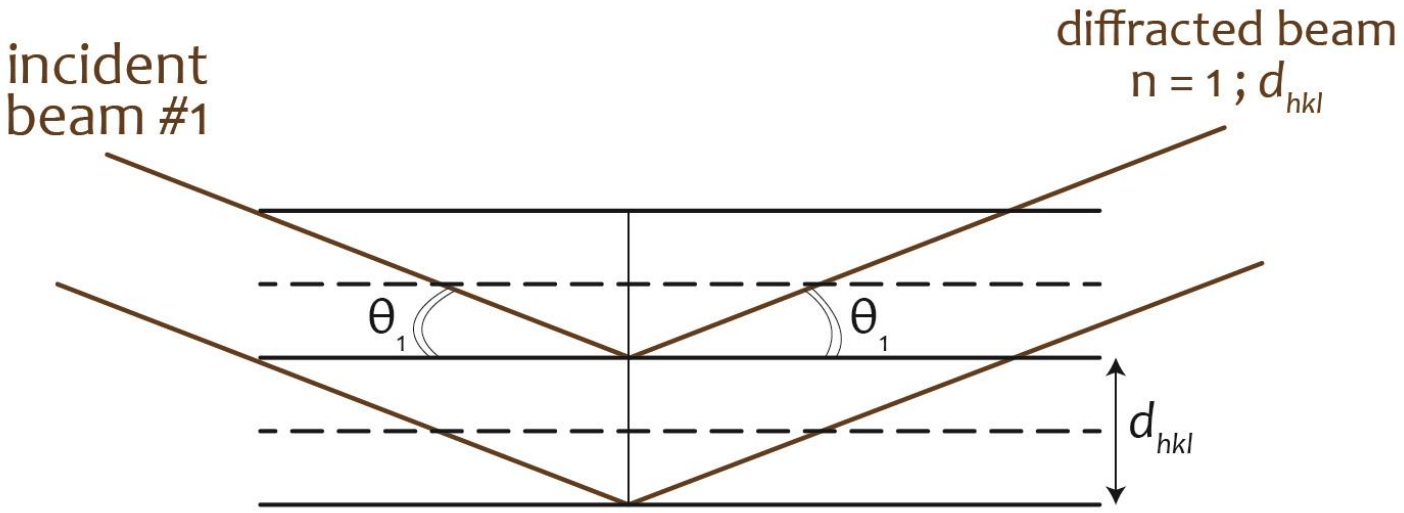


Nobel prize in Physics 1915

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Diffraction from crystal lattice planes

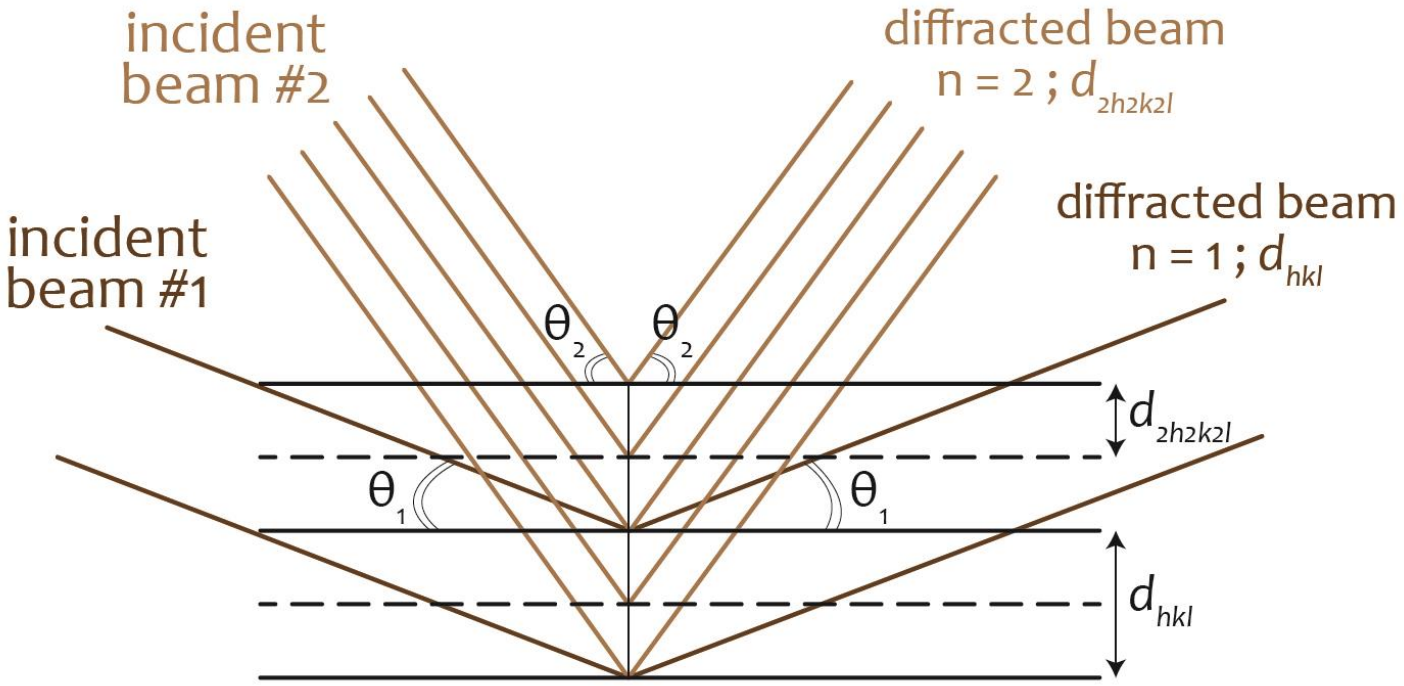
Relationship between diffraction order *n* and lattice planes.



$\longrightarrow 1\lambda = 2d_{hkl}.\sin(\theta_1)$

Diffraction from crystal lattice planes

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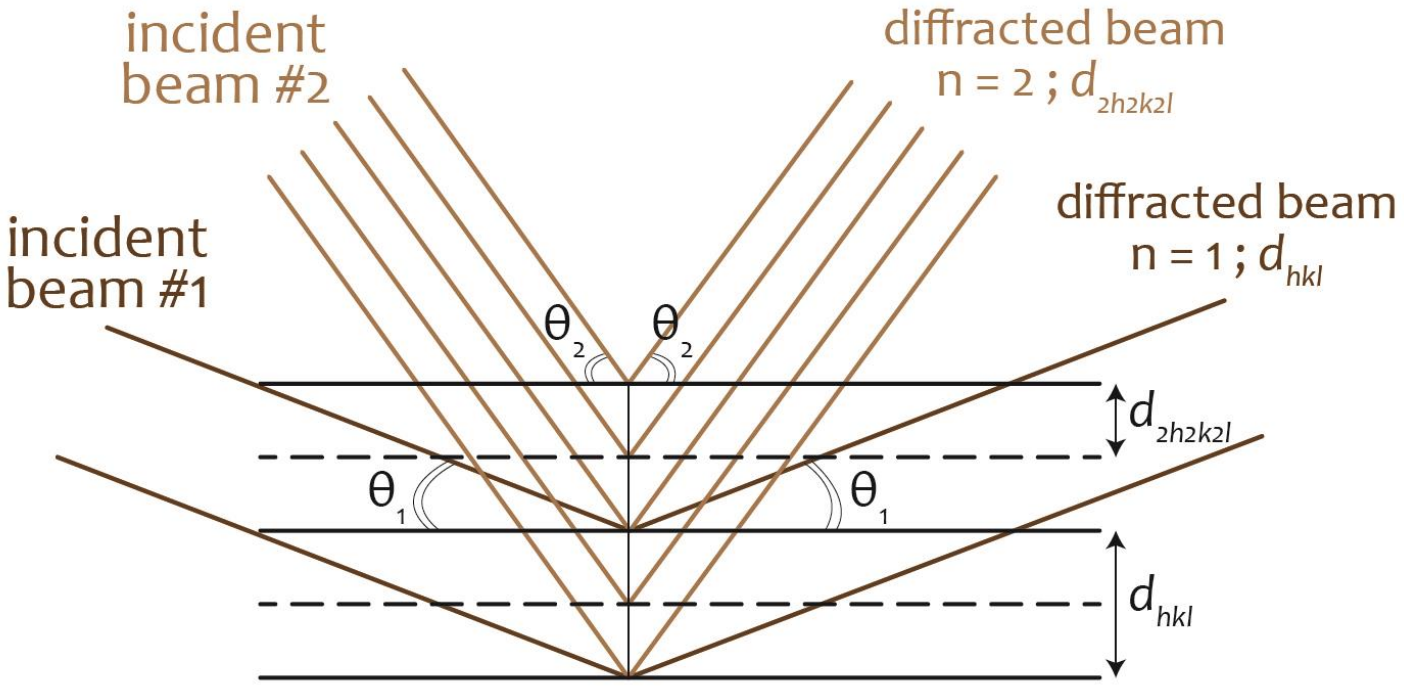


$\longrightarrow 1\lambda = 2d_{hkl}.\sin(\theta_1)$

$\longrightarrow 2\lambda = 2d_{2h2k2l}.\sin(\theta_2)$

Diffraction from crystal lattice planes

Relationship between diffraction order *n* and lattice planes.



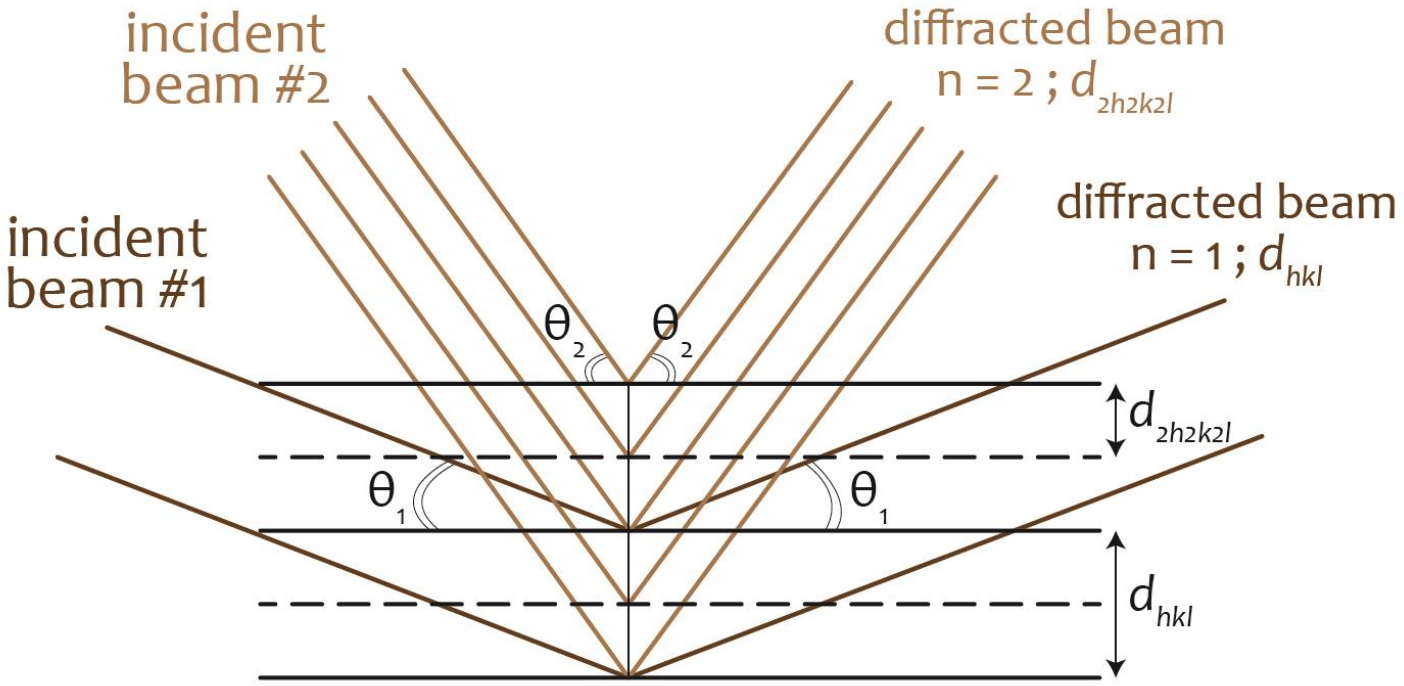
$\longrightarrow 1\lambda = 2d_{hkl}.\sin(\theta_1)$

$\longrightarrow 2\lambda = 2d_{2h2k2l}.\sin(\theta_2)$

$\longrightarrow n\lambda = 2d_{nhnknl}.\sin(\theta_n)$

Diffraction from crystal lattice planes

Relationship between diffraction order *n* and lattice planes.



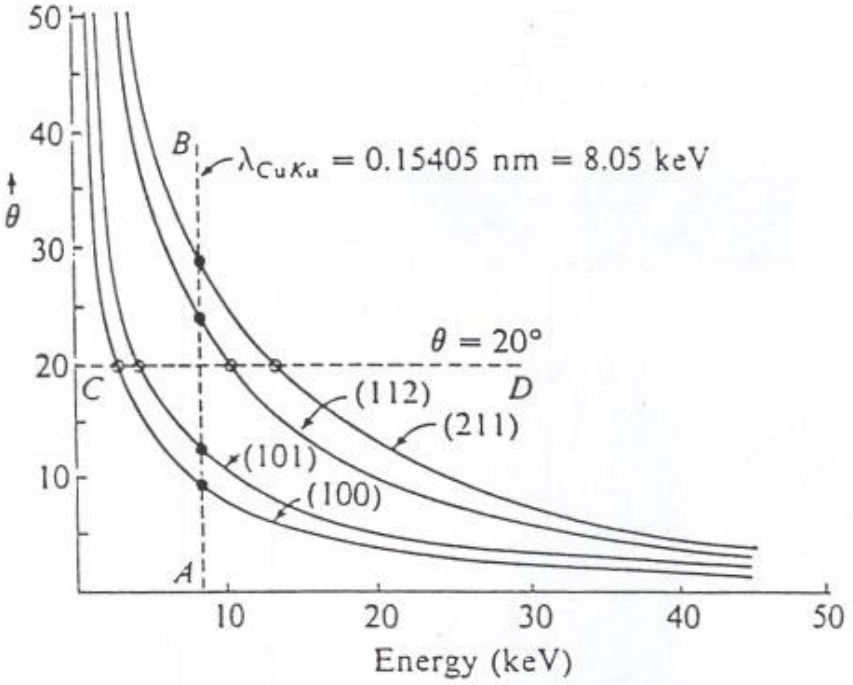


FIGURE 10.6 Plot of θ versus reciprocal wavelength $1/\lambda$ for low quartz according to the Bragg equation.

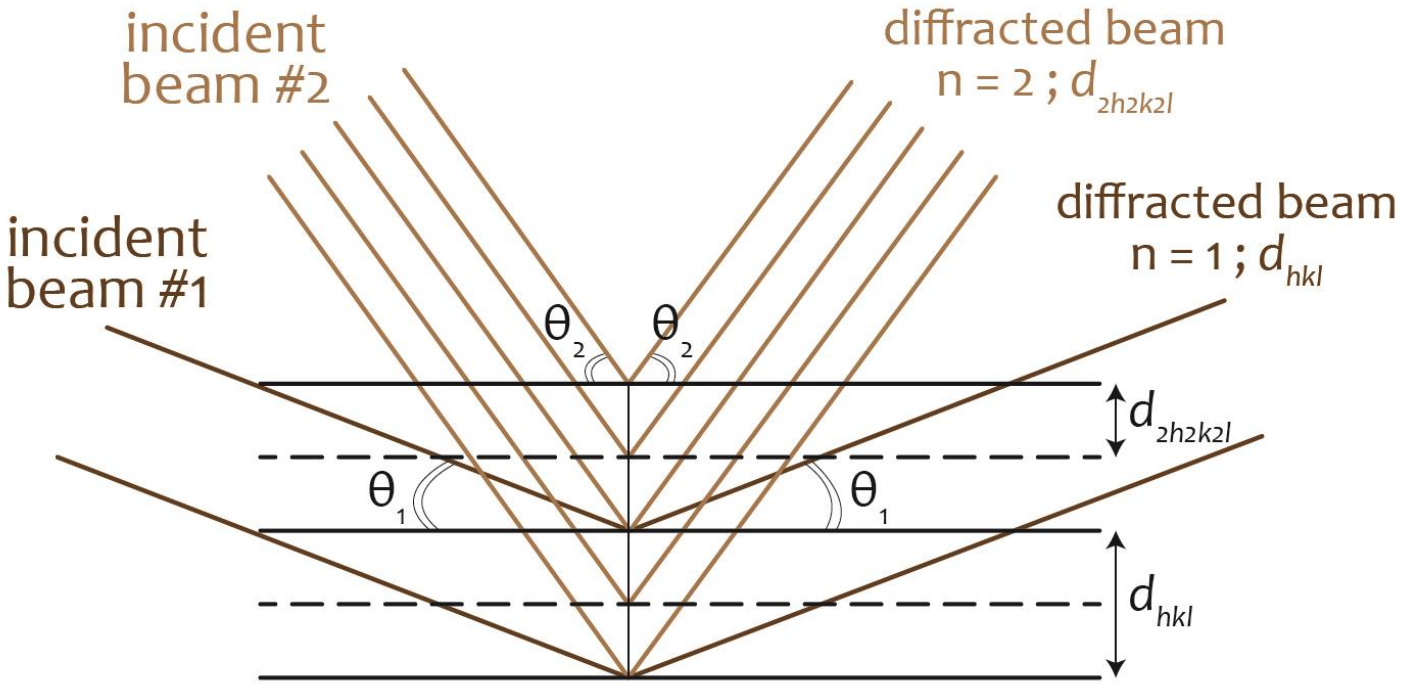
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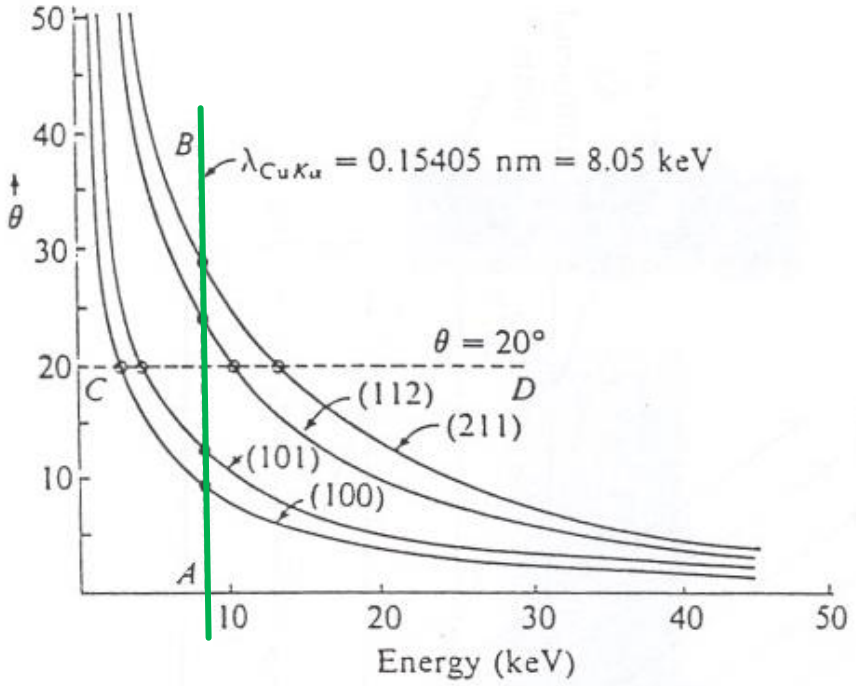


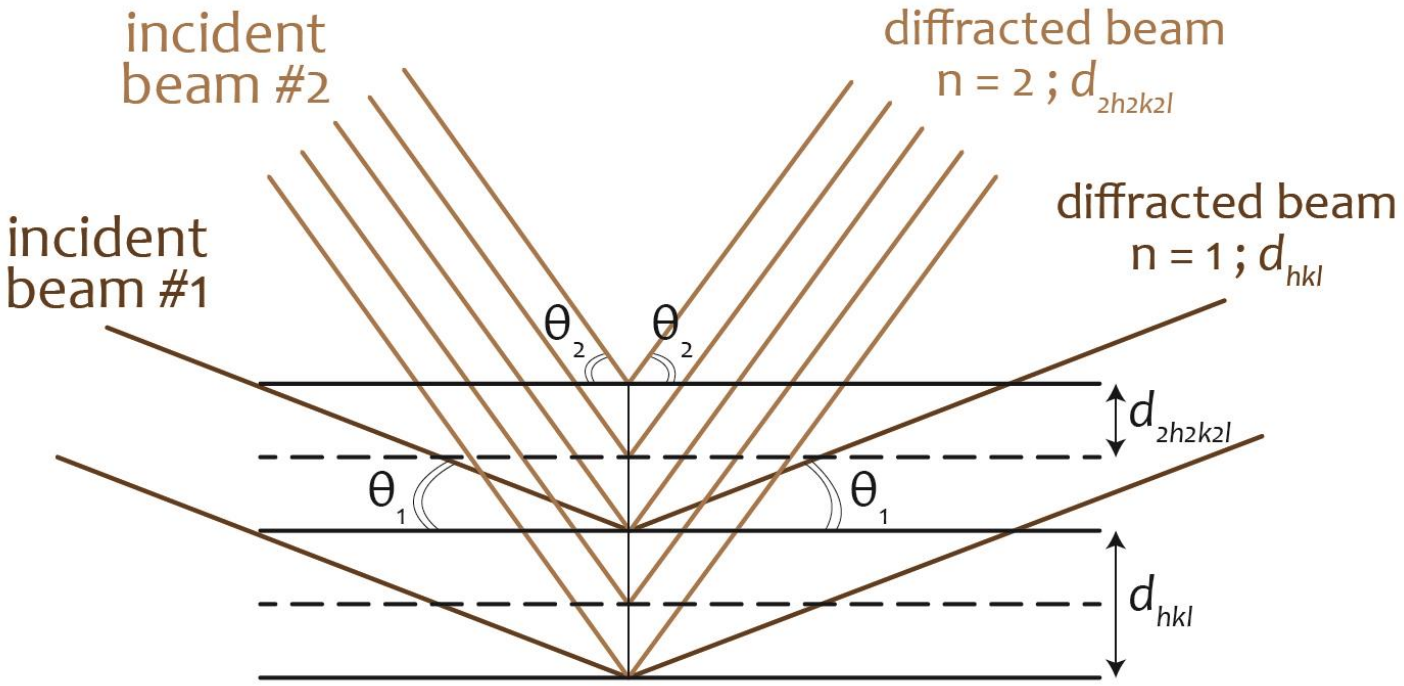
FIGURE 10.6 Plot of θ versus reciprocal wavelength $1/\lambda$ for low quartz according to the Bragg equation.

- With monochromatic beam

→ angularly dispersive XRD

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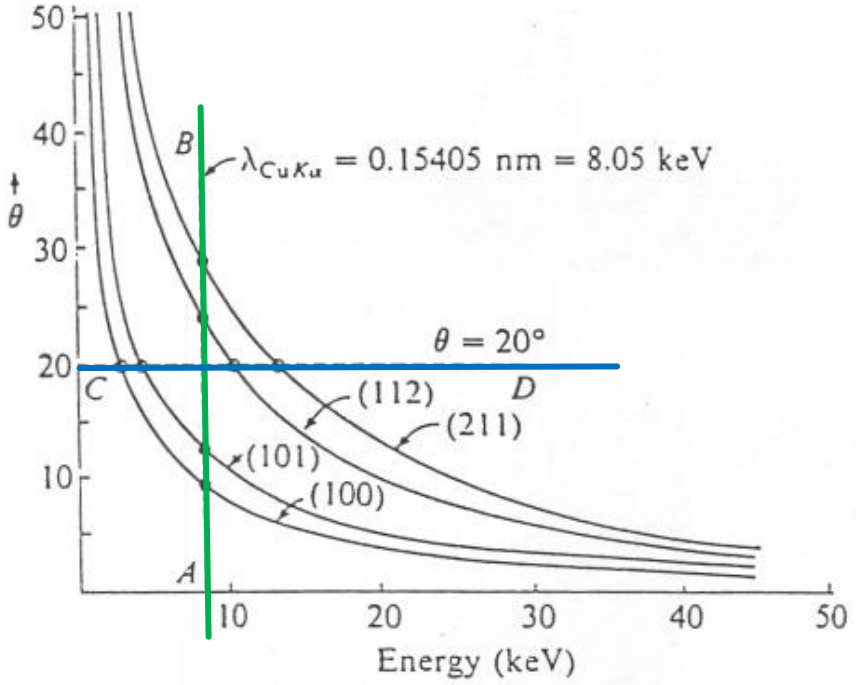
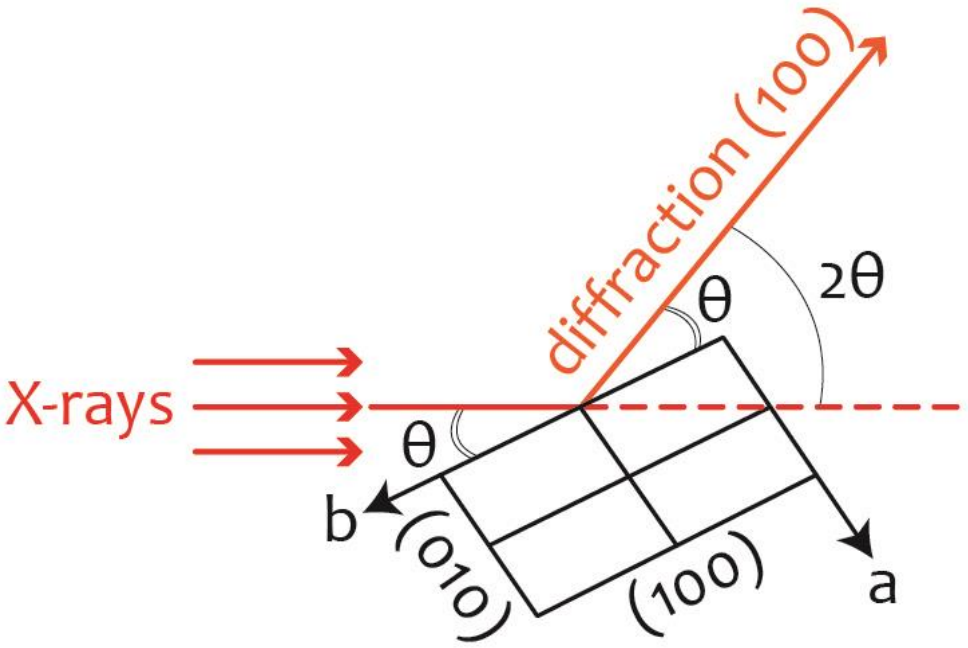


FIGURE 10.6 Plot of θ versus reciprocal wavelength $1/\lambda$ for low quartz according to the Bragg equation.

- With monochromatic beam
→ angularly dispersive XRD
- With white light (i.e. synchrotron)
→ energy dispersive XRD

Two main applications of X-ray diffraction

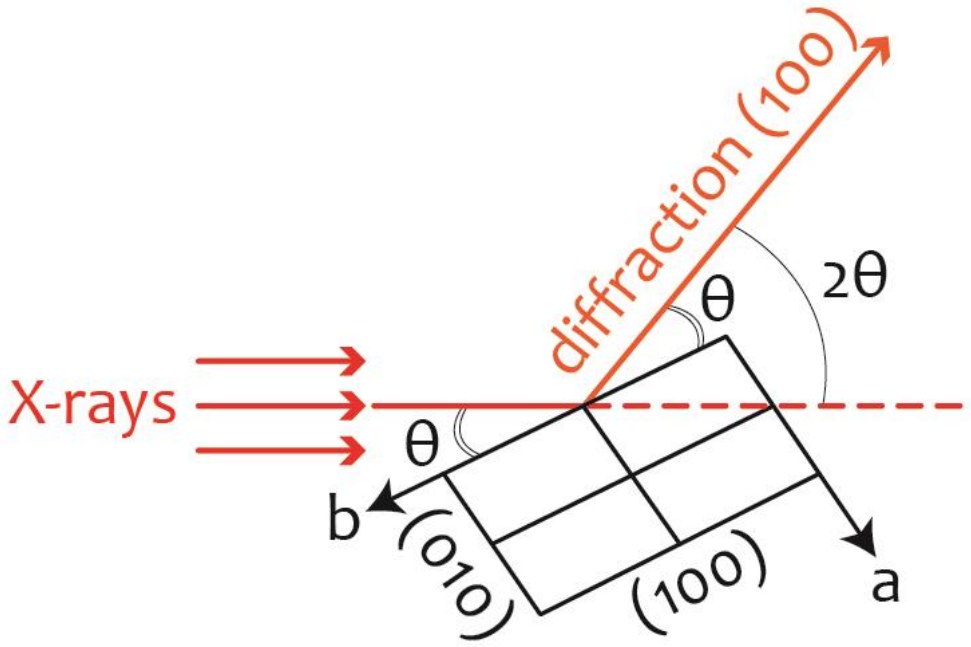


Single crystal X-ray diffraction

Detailed information about internal lattice of crystalline substances (mineral, molecules…).

ID of mineral from its X-ray diffraction pattern if all possible planes diffracted. Need to continuously rotate sample.

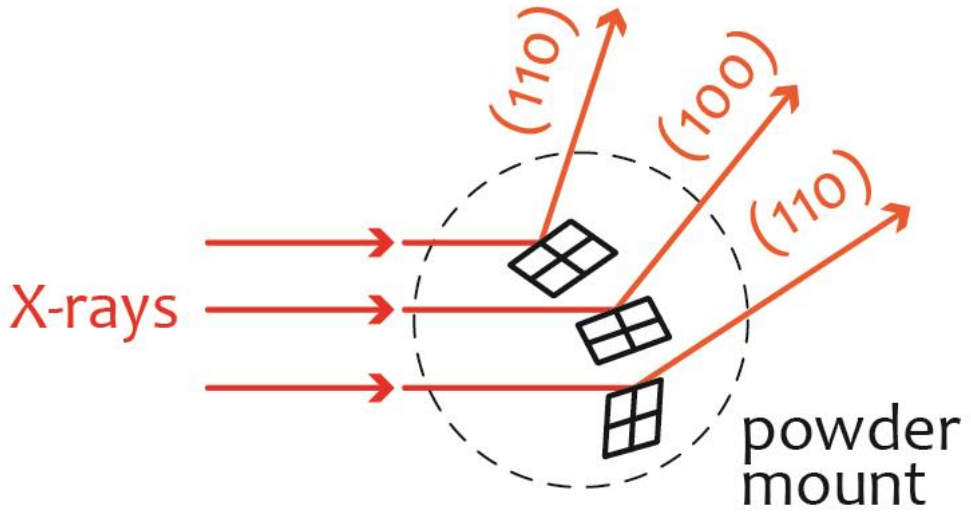
Two main applications of X-ray diffraction



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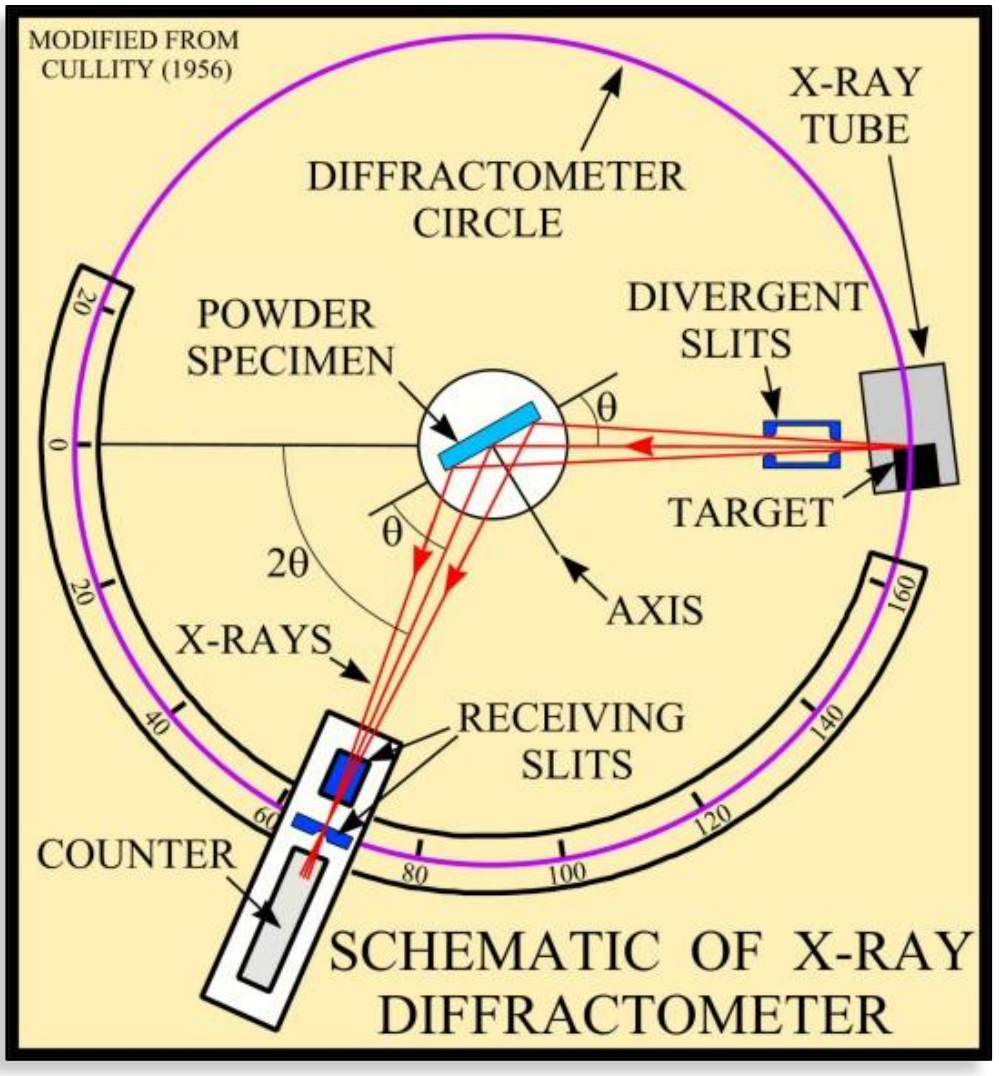
ID of mineral from its X-ray diffraction pattern if all possible planes diffracted. Need to continuously rotate sample.



Powder X-ray diffraction

Can analyse single crystal or rock powder. Assumes powder contains a lot of small fragments with random orientation. Can get full set of reflections without needing to rotate sample.

X-ray diffractometer



https://pubs.usgs.gov/of/2001/of01-041/htmldocs/xrpd.htm

- Powder contains millions of micro-crystals **randomly oriented** with respect to the incoming beam, which will give rise to **all possible planar diffraction** events at angles θ.
- In practice, we focus incident X-rays on the target through a Soller slit and **collect diffracted intensity** by rotating detector, so we record θ the incident angle and 2θ the diffracted angle.

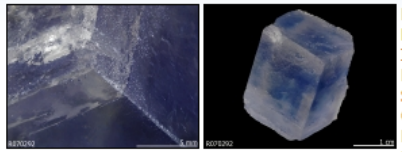
Example – halite (NaCl) powder

RRUFF Home | UA Mineralogy | Caltech Mineralogy | The IMA Mineral List | **Login**

RRUFF — Search **Sample Data** | Search **References** | **About** RRUFF | **Contact** Us

Halite R070292

Browse Search Results
<< Previous | Back to Search Results | Next >>
Record 1 of 3



Name: Halite
RRUFF ID: R070292
Ideal Chemistry: NaCl
Locality: Stassfurt, Stassfurt Deposit, Saxony-Anhalt, Germany
Source: Michael Scott S104647 [view label]
Owner: RRUFF
Description: Blue to colorless cube
Status: The identification of this mineral has been confirmed by X-ray diffraction.

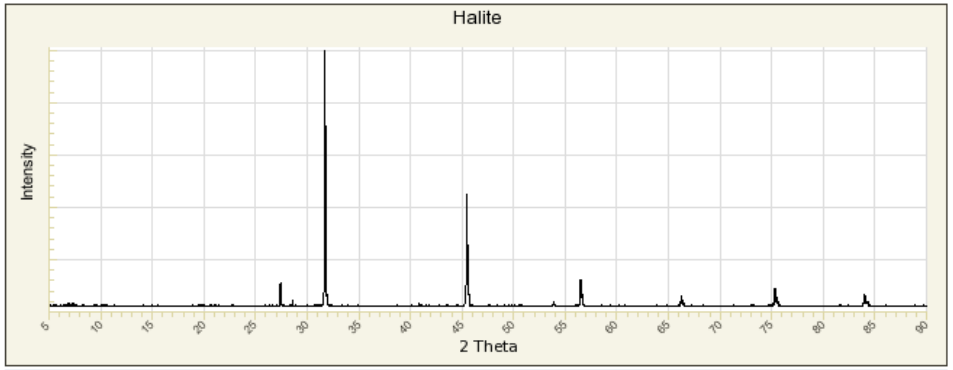
Mineral Group: [Rocksalt (15)]
Quick search: [All Halite samples (3)]

POWDER DIFFRACTION

RRUFF ID: R070292.1
Sample Description: Powder
Cell Refinement Output: **a:** 5.64167(4)Å **b:** 5.64167(4)Å **c:** 5.64167(4)Å
alpha: 90° **beta:** 90° **gamma:** 90° **Volume:** 179.565(4)Å^3 **Crystal System:** cubic

DOWNLOADS:

- Cell Refinement Data
- Cell Refinement Output Data
- DIF File
- X-ray Data (XY - Processed) RRUFF File
- X-ray Data (XY - RAW) RRUFF File

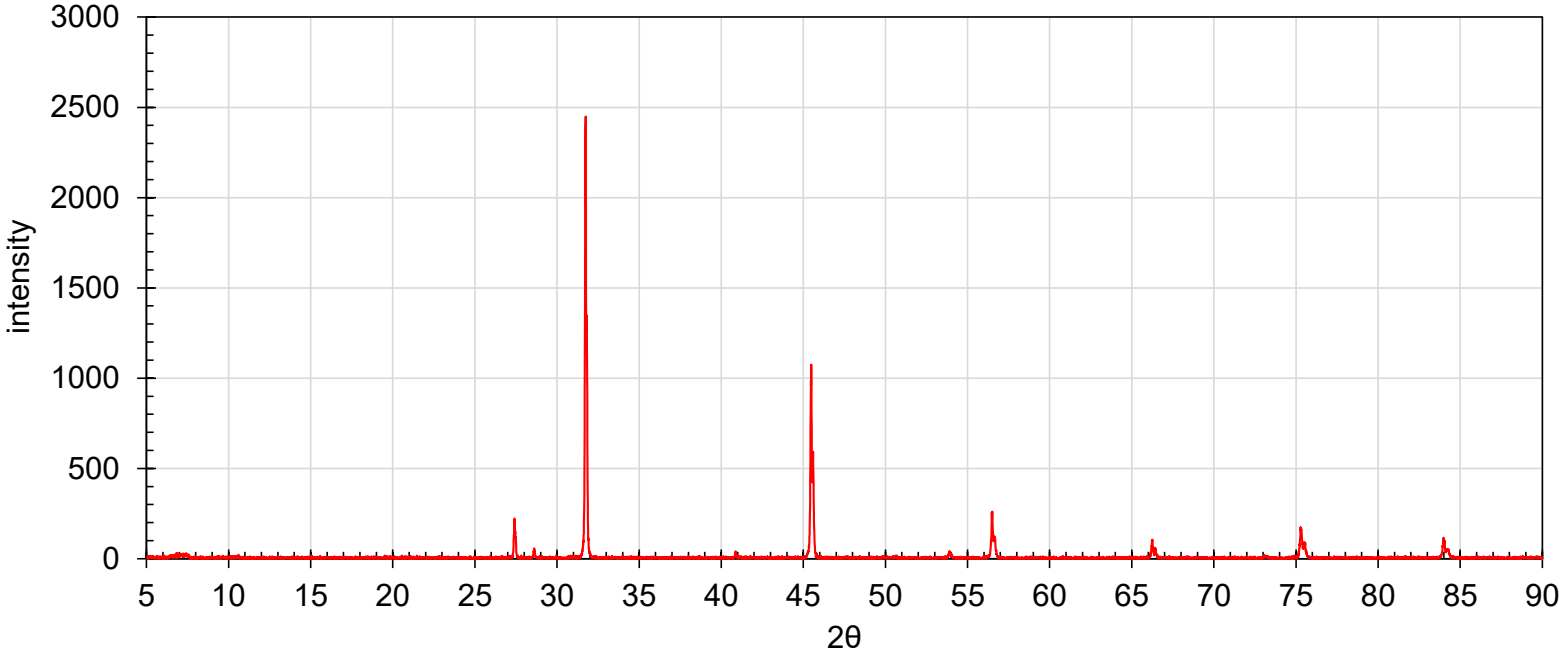


X Min: 5 X Max: 90 X Sort: asc refresh reset

https://rruff.info/halite/display=default/R070292

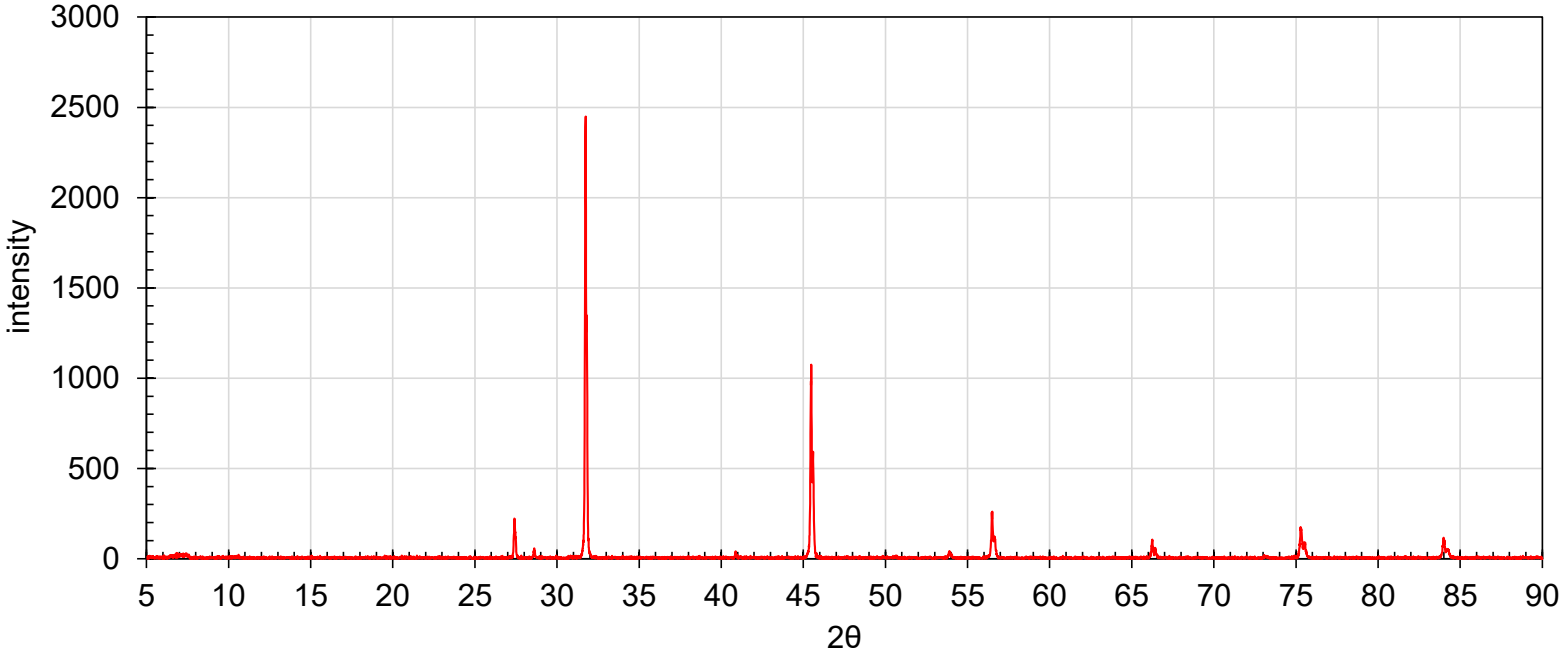
Example – halite (NaCl) powder

→ download and plot data



Example – halite (NaCl) powder

→ download and plot data



→ lookup DIF file

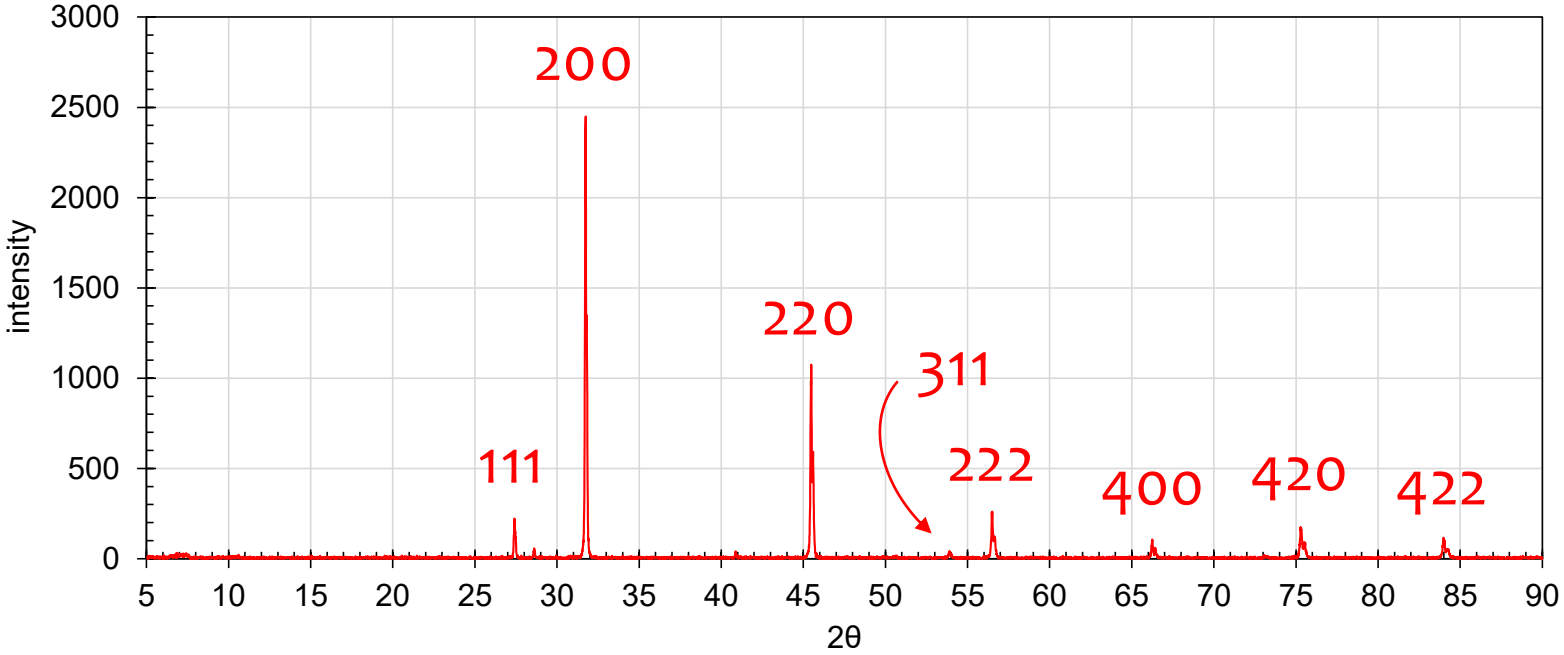
```
      Halite
Diffraction data computed using the structure from the paper listed below,
along with the cell parameters refined from the powder pattern for R070292
      Walker D, Verma P K, Cranswick L M D, Jones R L, Clark S M, Buhre S
      American Mineralogist 89 (2004) 204-210
      Halite-sylvite thermoelasticity
      Sample: msl416031, T = 25 C, P = 0.0 kbar, cell volume = 179.42 ang**3

      CELL PARAMETERS:   5.641667   5.641667   5.641667   90.00000   90.00000   90.00000
      SPACE GROUP: Fm3m
      X-RAY WAVELENGTH:     1.541838
              2-THETA      INTENSITY    D-SPACING   H   K   L
                27.40           8.77       3.2572   1   1   1
                31.73         100.00       2.8208   2   0   0
                45.47          61.00       1.9946   2   2   0
                53.89           2.21       1.7010   3   1   1
                56.49          18.23       1.6286   2   2   2
                66.24           7.42       1.4104   4   0   0
                73.08           1.06       1.2943   3   3   1
                75.31          18.39       1.2615   4   2   0
                84.00          12.69       1.1516   4   2   2
                84.00          12.69       1.1516   4   2   2
       XPOW Copyright 1993 Bob Downs, Ranjini Swaminathan and Kurt Bartelmehs
 For reference, see Downs et al. (1993) American Mineralogist 78, 1104-1107.
```

Bragg's law → $n\lambda = 2d.\sin\theta$

Example – halite (NaCl) powder

→ download and plot data



→ lookup DIF file

```
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Diffraction data computed using the structure from the paper listed below,
along with the cell parameters refined from the powder pattern for R070292
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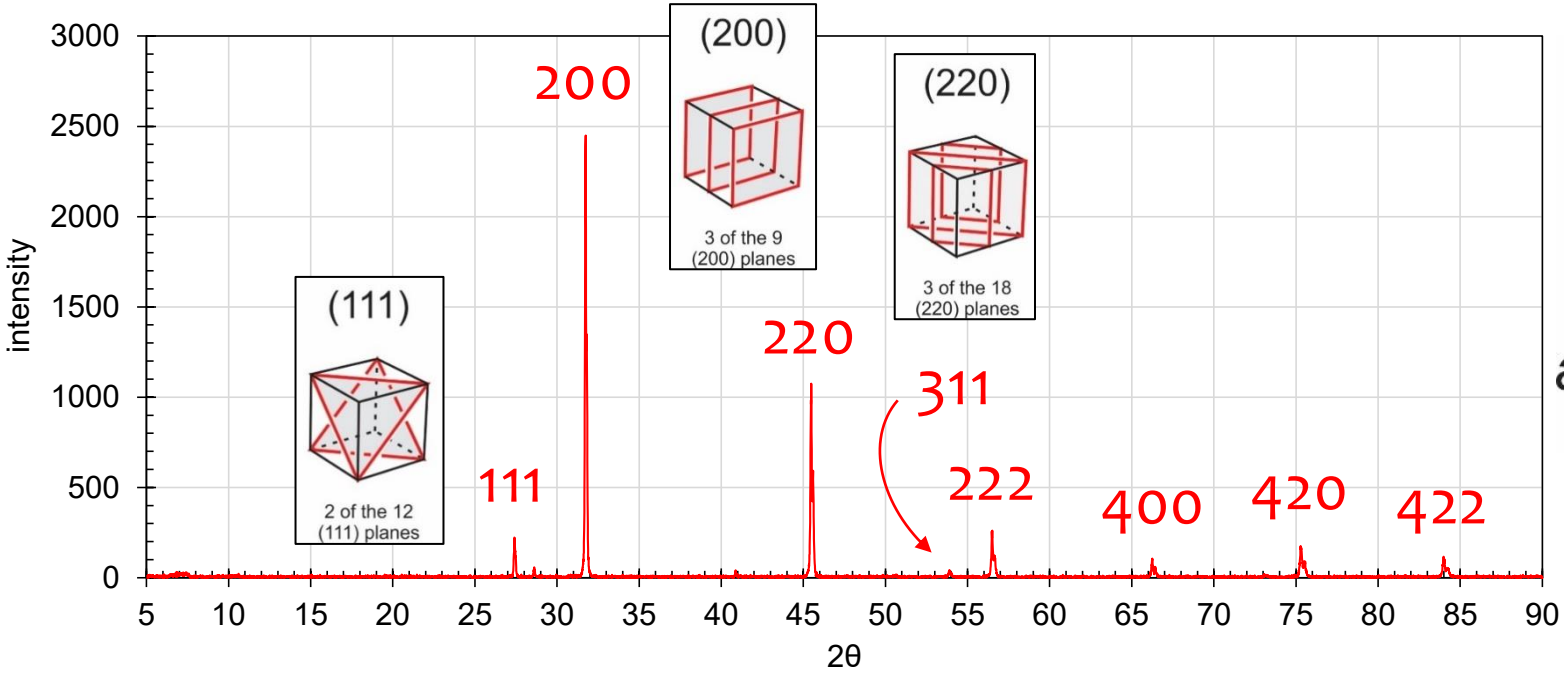
      CELL PARAMETERS:   5.641667   5.641667   5.641667   90.00000   90.00000   90.00000
      SPACE GROUP: Fm3m
      X-RAY WAVELENGTH:     1.541838
            2-THETA      INTENSITY    D-SPACING   H   K   L
              27.40          8.77       3.2572    1   1   1
              31.73        100.00       2.8208    2   0   0
              45.47         61.00       1.9946    2   2   0
              53.89          2.21       1.7010    3   1   1
              56.49         18.23       1.6286    2   2   2
              66.24          7.42       1.4104    4   0   0
              73.08          1.06       1.2943    3   3   1
              75.31         18.39       1.2615    4   2   0
              84.00         12.69       1.1516    4   2   2
              84.00         12.69       1.1516    4   2   2
       XPOW Copyright 1993 Bob Downs, Ranjini Swaminathan and Kurt Bartelmehs
 For reference, see Downs et al. (1993) American Mineralogist 78, 1104-1107.
```

→ identify peaks; *they correspond to diffraction on different lattice planes.*

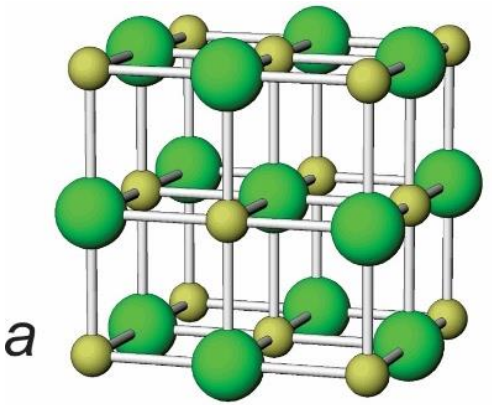
Bragg's law → $n\lambda = 2d.\sin\theta$

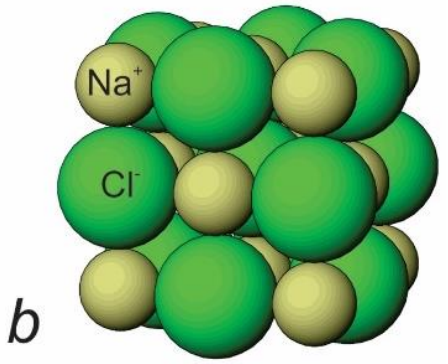
Example – halite (NaCl) powder

→ download and plot data



→ lookup DIF file





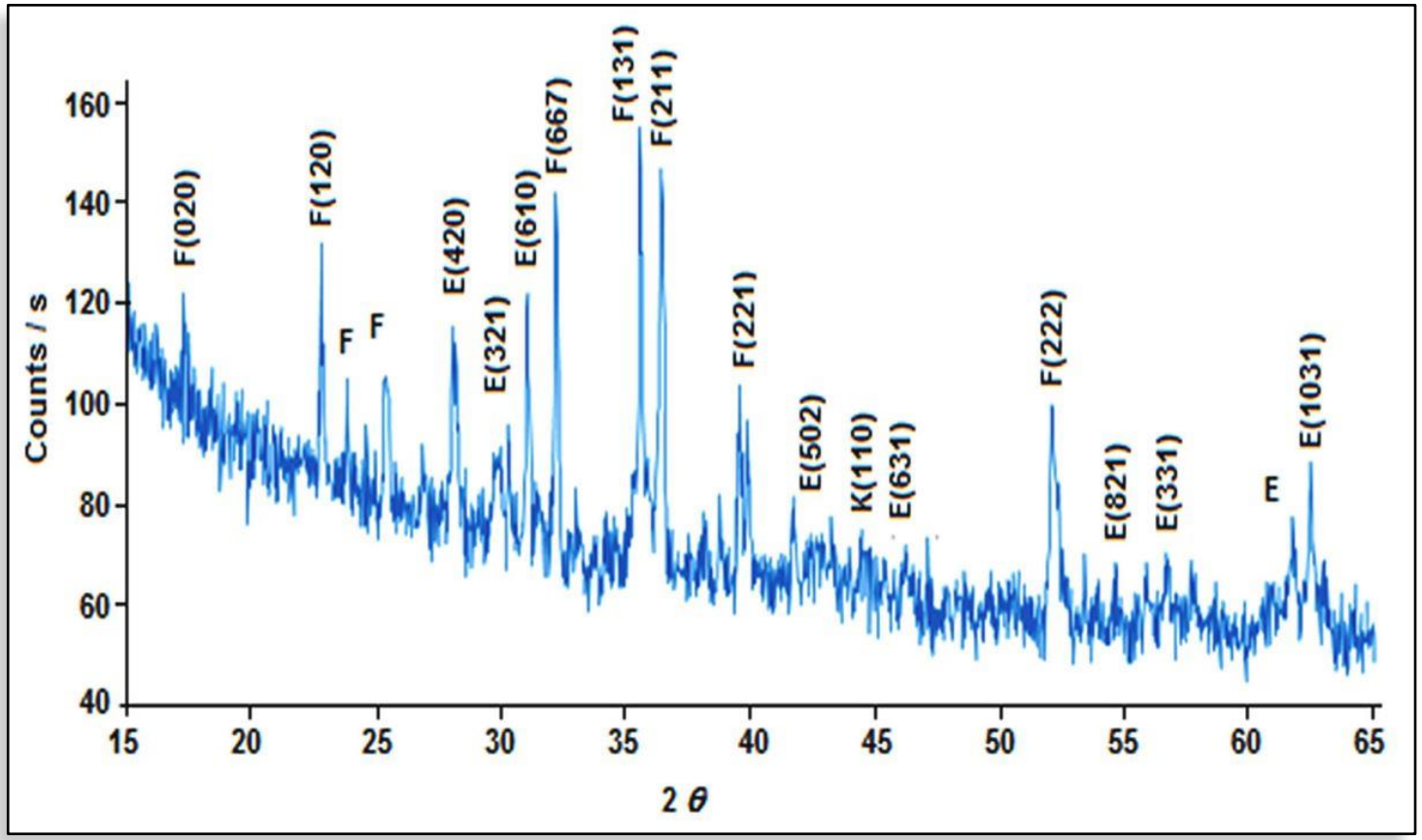

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66.24          7.42          1.4104     4   0   0
73.08          1.06          1.2943     3   3   1
75.31         18.39          1.2615     4   2   0
84.00         12.69          1.1516     4   2   2
84.00         12.69          1.1516     4   2   2
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For reference, see Downs et al. (1993) American Mineralogist 78, 1104-1107.
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https://opengeology.org/Mineralogy/12-x-ray-diffraction-and-mineral-analysis/

Example – Kamargaon meteorite powder

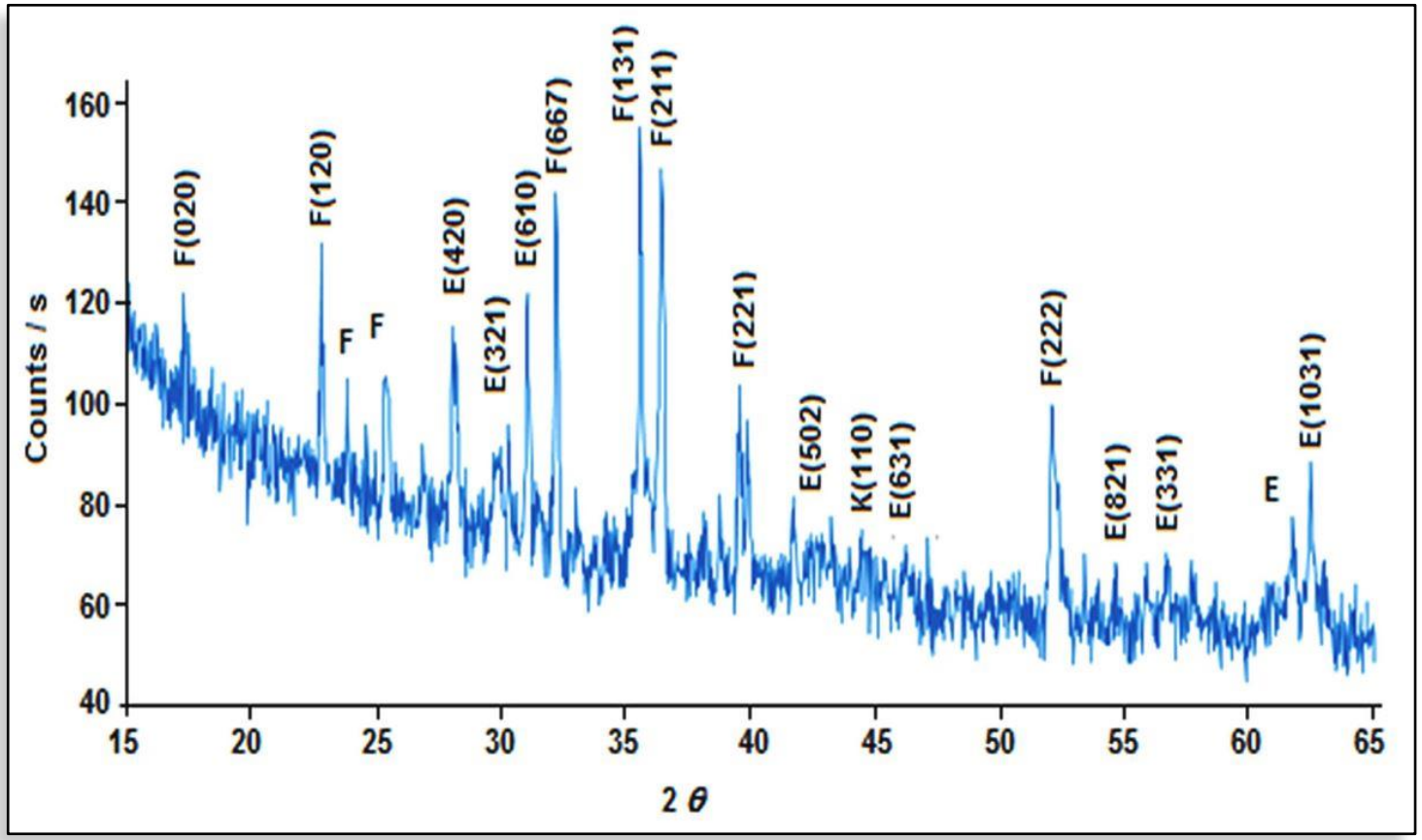


X-ray diffraction on powdered samples, using Cu Kα radiation (λKα ~ 1.54 Å), 2θ range of 15° to 65°, step size of 0.02° 2θ, and 1.2 sec / step.

Allows identifying main phases in the meteorite, which are olivine (forsterite - F), pyroxene (enstatite - E), and iron (kamacite - K).

From Saikia et al. (2017) Spectroscopy and Mineralogy of a Fresh Meteorite Fall Kamargaon (L6) Chondrite. Proc Indian Natn Sci Acad 83, 941-948.

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We'll work through an XRD dataset in the practical.